

Bell Inequalities Revisited Under Phase Admissibility

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Abstract

We rederive Bell inequalities within the framework of Phase Theory and demonstrate precisely which assumptions fail when global phase admissibility governs physical reality. Bell inequalities are shown to rely on an implicit factorizability assumption that is strictly incompatible with admissible phase topology. Phase Theory violates Bell inequalities without invoking hidden variables, superluminal influence, contextual collapse, or many-world branching. We prove that Bell inequality violation is a *necessary consequence* of global admissibility constraints, and show that all experimental results traditionally attributed to "quantum nonlocality" are recovered as structural features of phase-consistent reality.

We further prove the no-signaling theorem from first principles within

Phase Theory (establishing it as a theorem rather than a postulate), derive the Tsirelson bound from admissibility geometry, demonstrate the impossibility of super-quantum Popescu-Rohrlich correlations, and establish the Separability Failure Theorem showing that factorizability is strictly inconsistent with admissible phase topology for topologically non-trivial configurations. The joint admissible configuration space is shown to fail to decompose as a Cartesian product for entangled subsystems, yielding Bell violation as a topological inevitability rather than a dynamical accident.

The framework is compared systematically with Bohmian mechanics, many-worlds interpretation, GRW spontaneous collapse models, relational quantum mechanics, and QBism, with a comprehensive comparison table enumerating the ontological and predictive differences. Eight classes of falsifiable predictions are enumerated with specificity, and experimental protocols for testing Phase Theory's distinctive Bell-type claims are specified. We conclude that Bell's theorem does not establish the nonlocality of nature but rather the failure of separability as an ontological principle — and that global phase admissibility is the correct structural explanation for the full body of experimental evidence on correlated quantum systems.

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1. Why Bell Must Be Revisited

Bell's theorem occupies a singular position in the history of physics. Derived in 1964 by John Stewart Bell as a mathematical sharpening of the Einstein-Podolsky-Rosen (EPR) puzzle, it has since been elevated to a kind of epistemic landmark: the result that, more than any other, is said to establish that the universe is irreducibly strange, that classical intuitions about locality and reality must be abandoned, and that quantum mechanics describes something fundamentally beyond the reach of any ordinary physical picture. The decades of experimental work that followed — culminating in the loophole-free Bell tests of 2015 — have only deepened this impression. Today, textbooks routinely present Bell's theorem as proof that either *locality* fails or *realism* fails, with many presentations declaring that no local realistic theory can reproduce quantum mechanical predictions. The theorem is treated not merely as a mathematical result but as a settled verdict on the deep structure of nature.

This paper argues that this verdict is premature — and that it rests on a set of ontological assumptions that are neither necessary nor universally shared. Specifically, we contend that the standard conclusions drawn from Bell's theorem presuppose an ontological framework — one in which physical systems have locally definite states, in which the joint probability distributions over outcomes factor into independent local contributions, and in which the structure of physical reality is fundamentally separable — that Phase Theory explicitly and systematically rejects. Bell's mathematics is impeccably correct. His standard interpretation, however, is not the only interpretation consistent with the mathematics, and it is not the interpretation forced upon us by the experimental evidence.

The standard reading of Bell's theorem holds that it establishes one of three conclusions: (1) locality fails, meaning that spacelike-separated events can exert direct causal influence on one another; (2) realism fails, meaning that

physical systems do not possess definite properties prior to measurement; or (3) hidden variables must be nonlocal, meaning that any completion of quantum mechanics must involve instantaneous action at a distance. Phase Theory rejects all three conclusions — not because it disputes the experimental evidence, which it reproduces exactly, but because it rejects the ontological framework from which these three conclusions are derived. The framework Phase Theory rejects is the assumption of *ontological separability*: the assumption that the physical state of a composite system is, in principle, decomposable into independent states of its subsystems.

This paper makes the following main contributions. First, we give a precise formal account of why factorizability — the mathematical expression of separability — fails in Phase Theory, and we prove that this failure is not a contingent feature of the theory but a structural necessity of global admissibility (Section 12). Second, we derive the CHSH violation and Tsirelson bound from admissibility geometry without invoking quantum mechanical formalism (Sections 13–15). Third, we prove no-signaling as a theorem, not a postulate, within Phase Theory (Section 17). Fourth, we establish the Separability Failure Theorem, which formalizes the incompatibility of factorizability with globally admissible phase topology (Section 24). Fifth, we compare Phase Theory systematically with all major alternative interpretations and show that it achieves Bell violation through a mechanism distinct from all of them (Sections 29–33). Sixth, we enumerate eight classes of falsifiable experimental predictions (Section 37).

The scientific importance of this revisitation cannot be overstated. Bell's theorem is the linchpin of quantum foundations; any theory that proposes a new ontological framework must give a detailed and rigorous account of how it handles Bell's result. Providing that account is the primary purpose of this paper.

2. Historical Context of Bell's Theorem

The story begins in 1935 with the paper by Albert Einstein, Boris Podolsky, and Nathan Rosen, universally known by the initials EPR [1]. Einstein, Podolsky, and Rosen constructed a thought experiment involving two quantum particles prepared in a correlated state and then separated to a large distance. By measuring one particle, one could infer the state of the other without disturbing it, suggesting — by EPR's criterion of physical reality — that the second particle possessed a definite property prior to and independent of measurement. If quantum mechanics does not assign a definite value to this property prior to measurement, then, EPR concluded, quantum mechanics is incomplete: there must be additional variables, hidden from the quantum formalism, that determine the outcome.

Niels Bohr responded to EPR in the same year, arguing that the EPR criterion of reality was inapplicable because the act of measuring one particle, even at a distance, affected the legitimate description of the experimental context within which the other particle's property was defined [2]. This exchange defined the landscape of the foundations debate for nearly three decades, but it remained at the level of philosophical argumentation. Neither side had identified a mathematically precise criterion by which the two positions could be experimentally distinguished.

That criterion arrived in 1964 with Bell's landmark paper "On the Einstein Podolsky Rosen Paradox" [3]. Bell showed that any theory satisfying what he called *local causality* — roughly, the condition that the outcome at one measurement site is independent of the setting at the other, given a complete specification of the local state — must satisfy an inequality involving measurable correlation functions. Quantum mechanics predicts violations of this inequality for appropriately entangled states and appropriately chosen measurement settings. Bell had converted a philosophical dispute into an empirical question.

The practical implementation of Bell's test was given a more experimentally accessible form in 1969 by Clauser, Horne, Shimony, and Holt (CHSH), who introduced the now-standard CHSH inequality [4]. The CHSH formulation uses four correlation functions corresponding to two measurement settings at each of two sites and produces an inequality whose classical bound is 2 and whose quantum mechanical maximum violation is $2\sqrt{2}$ — a value known as the Tsirelson bound [5].

The first experimental test was performed by Freedman and Clauser in 1972 [6], finding a violation of the Bell inequality consistent with quantum predictions. The landmark series of experiments by Aspect, Grangier, and Roger in 1981–1982 [7, 8] substantially strengthened the case, employing rapidly switched analyzers to close the locality loophole and reporting clear violations consistent with quantum mechanics. Further experiments by Weihs et al. in 1998 [9] used electronically switched measurement settings to implement strict Einstein locality conditions. Rowe et al. in 2001 [10] achieved nearly perfect detection efficiency with trapped ions, closing the detection loophole for the first time.

The year 2015 was decisive. Three independent groups — Hensen et al. [11], Giustina et al. [12], and Shalm et al. [13] — published loophole-free Bell inequality violation experiments, simultaneously closing both the locality and detection loopholes. The Big Bell Test of 2018 [14] used human-generated measurement choices, drawn from over 100,000 participants worldwide, to additionally address the freedom-of-choice loophole. These experiments constitute overwhelming empirical confirmation that quantum mechanical predictions for correlated systems are correct and that the Bell inequalities are violated by nature.

The current consensus, shared by virtually all working physicists, is that quantum mechanics correctly describes these phenomena and that no local realistic hidden variable theory can reproduce the experimental results.

However, this consensus addresses the empirical question decisively while leaving the interpretive question entirely open. The experimental data tell us what the correlations are; they do not, by themselves, tell us *why* the correlations take the form they do or what ontological story explains them. It is precisely this explanatory gap that Phase Theory proposes to fill — without invoking nonlocal causation, hidden variables, or any ontology beyond globally admissible phase configurations.

3. The EPR Argument and Its Assumptions

The EPR argument proceeds from two foundational premises. The first is the *completeness criterion*: a physical theory is complete if and only if every element of physical reality has a counterpart in the theory. The second is the *criterion of physical reality*: if, without in any way disturbing a system, one can predict with certainty the value of a physical quantity, then there exists an element of physical reality corresponding to that quantity.

Definition 3.1 (EPR Completeness).

A physical theory T is *complete* if and only if for every element of physical reality E , there exists a counterpart element $e \in T$ such that e represents E exactly.

Definition 3.2 (EPR Reality Criterion).

A physical quantity Q has an element of physical reality if and only if its value can be predicted with probability 1 without disturbing the system to which Q pertains.

From these two premises, EPR construct the following argument. Consider two particles prepared in the singlet state and separated to a large distance. Measuring the spin of particle 1 along direction $\hat{n}$ with result $+1$ allows one to predict with certainty that measuring particle 2 along the same direction will yield -1 . By Definition 3.2, there is therefore an element of physical reality corresponding to the spin of particle 2 along direction $\hat{n}$, determined prior to and independent of measurement. But quantum mechanics does not assign a definite spin value to particle 2 prior to measurement; hence, by Definition 3.1, quantum mechanics is incomplete.

The EPR argument thus presents a dilemma: either quantum mechanics is incomplete (and there exist hidden variables that restore determinism and completeness), or quantum mechanics is complete (and it must describe a genuine nonlocal action at a distance, since the measurement of particle 1 instantaneously affects what can be said about particle 2, regardless of their spatial separation). Einstein regarded the second horn as unacceptable — a violation of what he called *local causality* or the principle of separability — and therefore concluded that hidden variables must exist.

Phase Theory dissolves this dilemma entirely by rejecting its presupposition structure. The EPR argument presupposes that the physical state of a composite system must, at the ontological level, decompose into the states of its components. This presupposition — which EPR treat as self-evident — is precisely what Phase Theory denies. In Phase Theory, the fundamental ontological entity is the global admissible phase configuration of the whole system, and this global configuration cannot in general be decomposed into independent local states of the subsystems. The EPR "element of reality" associated with particle 2's spin is not an independently existing property of particle 2; it is a structural feature of the global admissible configuration that includes both particles and their correlations. Measuring particle 1 does not *create* the property of particle 2; it reveals a constraint that was already

globally enforced by admissibility. This is not nonlocal causation — it is global structural coherence.

Neither horn of the EPR dilemma applies within Phase Theory. Quantum mechanics is not incomplete (no additional hidden variables are needed), nor does it describe nonlocal action at a distance (no causal influence travels between the measurement sites). What the EPR argument misses is the third possibility: that physical reality is globally organized in a way that makes separability unavailable as an ontological primitive.

4. Bell's Formal Argument

We now present Bell's 1964 derivation in full formal detail, identifying each assumption with precision so that Phase Theory's stance toward each assumption can be made explicit in the following section.

Bell assumes the existence of a *hidden variable space* (Λ, μ) , where Λ is a measurable space and μ is a probability measure over Λ . Each element $\lambda \in \Lambda$ represents a complete specification of the physical state of the joint system at the moment of preparation. The hidden variables are assumed to carry all information relevant to determining the measurement outcomes.

Definition 4.1 (Outcome Functions).

For measurement settings a and b at sites A and B respectively, the outcome functions are $A(a, \lambda) \in \{-1, +1\}$ and $B(b, \lambda) \in \{-1, +1\}$, which assign definite outcomes to each hidden variable value $\lambda \in \Lambda$.

The two-particle correlation function for settings a and b is then defined as:

$$(Eq. 1) \ E(a, b) = \int_{\Lambda} A(a, \lambda) \cdot B(b, \lambda) \, d\mu(\lambda)$$

This expression encodes the assumption that the joint outcome is determined by the shared hidden variable λ , with the local outcomes $A(a, \lambda)$ and $B(b, \lambda)$ depending only on the local measurement setting and the shared variable.

For four correlation functions with settings a, a' at site A and b, b' at site B, Bell derives the following inequality by algebraic manipulation of the outcome functions:

$$(Eq. 2) |E(a, b) - E(a, b')| + |E(a', b) + E(a', b')| \leq 2$$

The derivation of (Eq. 2) proceeds as follows. Define $S(\lambda) = A(a, \lambda)[B(b, \lambda) - B(b', \lambda)] + A(a', \lambda)[B(b, \lambda) + B(b', \lambda)]$. Since each factor is ± 1 , either $B(b, \lambda) = B(b', \lambda)$, in which case $B(b, \lambda) - B(b', \lambda) = 0$ and $B(b, \lambda) + B(b', \lambda) = \pm 2$, giving $|S(\lambda)| = 2$; or $B(b, \lambda) \neq B(b', \lambda)$, in which case $B(b, \lambda) + B(b', \lambda) = 0$ and $|B(b, \lambda) - B(b', \lambda)| = 2$, again giving $|S(\lambda)| = 2$. Therefore $|S(\lambda)| \leq 2$ for all λ , and integrating over μ yields $|S| \leq 2$, which is precisely (Eq. 2).

The four assumptions implicit in this derivation are:

1. **Outcome Realism:** For each $\lambda \in \Lambda$, the outcome functions $A(a, \lambda)$ and $B(b, \lambda)$ are well-defined, single-valued functions yielding definite values in $\{-1, +1\}$.
2. **Statistical Independence (Freedom of Choice):** The distribution μ over hidden variables is independent of the choice of measurement settings a and b ; equivalently, $P(a, b | \lambda) = P(a) \cdot P(b)$ for all $\lambda \in \Lambda$.
3. **Local Causality:** The outcome at site A depends only on the local setting a and the shared hidden variable λ , not on the remote setting b ; and symmetrically for site B.
4. **Factorizability:** The joint probability distribution over outcomes factors as $P(a, b | \lambda) = P(a | \lambda) \cdot P(b | \lambda)$ — that is, conditional on the hidden variable, the outcomes at the two sites are statistically independent.

Of these four assumptions, we shall argue in the following section that *factorizability* is the load-bearing assumption whose failure under Phase Theory explains Bell inequality violation. The others are either accepted by Phase Theory or are independently motivated by considerations that do not depend on factorizability.

5. What Bell Inequalities Actually Assume

A careful analysis reveals that the four assumptions of Section 4 are not equally fundamental and are not equally well-motivated. We examine each in turn and specify Phase Theory's stance toward each.

Outcome Realism. Bell's derivation requires that $A(a, \lambda)$ and $B(b, \lambda)$ be well-defined single-valued functions. This amounts to the assumption that, for each complete hidden variable specification λ , there exist determinate outcomes prior to measurement. Phase Theory does not accept this assumption in the required form. In Phase Theory, outcomes are not pre-assigned properties of subsystems; they are consistent selections from globally admissible phase configurations. A measurement event is not the revelation of a pre-existing value but the identification of a globally coherent outcome assignment. This is ontologically distinct from both the claim that outcomes are random (as in standard quantum mechanics) and from the claim that they are predetermined (as in hidden variable theories).

Statistical Independence. Phase Theory accepts statistical independence in full. The measurement settings chosen by experimenters are not determined by the phase substrate; they are freely chosen, and Phase Theory's admissibility constraints do not depend on which settings are chosen prior to the measurement event. Proposition 16.1 will establish this formally. Phase Theory is not superdeterministic.

Local Causality. Phase Theory accepts local causality in the sense that no causal influence propagates faster than light between the measurement sites. Bell violations in Phase Theory are not the result of any subluminal or superluminal causal link. The no-signaling theorem (Theorem 17.1) will establish this formally.

Factorizability. Phase Theory rejects factorizability. This is the crucial point. Factorizability — the condition $P(a, b \mid \lambda) = P(a \mid \lambda) \cdot P(b \mid \lambda)$ — is not a consequence of local causality alone; it is an additional assumption encoding *ontological separability*. A system whose joint probability distribution is determined by a globally admissible constraint on a non-separable configuration space will generically fail to be factorizable even when no superluminal signals are sent. The distinction between local causality and factorizability is the central conceptual lesson of this paper.

Theorem 5.1 (Factorizability Implies Separability).

If the joint probability distribution $P(a, b \mid \lambda)$ is factorizable for all $\lambda \in \Lambda$, then the joint outcome space can be written as a Cartesian product of independent local outcome spaces: $O_{AB} = O_A \times O_B$. Equivalently, factorizability implies ontological separability. However, the converse does not hold: local causality (the absence of superluminal signals) does not imply factorizability.

Proof. If $P(a, b \mid \lambda) = P(a \mid \lambda) \cdot P(b \mid \lambda)$ for all λ , then the joint measure decomposes as the product of the marginals for every conditioning value λ . This means the joint outcome is determined by two independent random variables (one at each site), which is precisely the condition for the outcome space to be a Cartesian product $O_A \times O_B$. For the non-converse: a system with global admissibility constraints can have spacelike-separated sites with

no causal influence between them (local causality satisfied) while still having a joint distribution determined by a non-factorizable global constraint. The no-signaling theorem (Theorem 17.1) provides the explicit construction. $\square$

The consequence of Theorem 5.1 is that rejecting factorizability is *not* equivalent to rejecting locality. One can maintain full local causality — no superluminal signaling, no causal influence between spacelike-separated events — while also rejecting factorizability, provided that the joint probability distribution is determined by a globally admissible constraint that cannot be decomposed into local factors. This is precisely the situation in Phase Theory, and it is the conceptual key to understanding how Bell inequality violation is possible without any form of nonlocal causation.

6. Factorizability as the Hidden Assumption

The purpose of this section is to isolate factorizability as the primary assumption driving Bell's inequality, to define it precisely, and to distinguish it carefully from the related but distinct concept of local causality.

Definition 6.1 (Factorizability).

A joint outcome distribution $P(a, b \mid \lambda)$ over measurement outcomes $a \in O_A$ and $b \in O_B$, conditioned on a parameter $\lambda \in \Lambda$, is *factorizable* if and only if:

$$(\text{Eq. 3}) \quad P(a, b \mid \lambda) = P(a \mid \lambda) \cdot P(b \mid \lambda) \quad \text{for all } \lambda \in \Lambda, a \in O_A, b \in O_B$$

Proposition 6.1 (Factorizability Encodes Separability).

Factorizability is not a consequence of local causality; it encodes a further ontological commitment to the *separability* of the physical state of the composite system — the assumption that the complete state of the composite system is fully specified by the pair of local states (λ_A, λ_B) , with no irreducible joint structure.

Proof. Local causality requires only that $A(a, \lambda)$ does not depend on b and $B(b, \lambda)$ does not depend on a — a condition about causal influence, not about the structure of the state space. Factorizability additionally requires that, given λ , the outcomes at the two sites are statistically independent. If λ is a global variable encoding joint correlations (as in Phase Theory, where λ encodes a global admissible configuration), then conditioning on λ does not in general render the outcomes independent, even if no causal influence is present. The factorizability condition (Eq. 3) fails precisely when the global variable λ encodes irreducible joint structure. $\square$

Proposition 6.2 (Globally Constrained Distributions Need Not Be Factorizable).

A joint probability distribution $P(a, b)$ arising from a global constraint C on a configuration space Φ is generically non-factorizable, even if C enforces no superluminal signals between sites A and B .

Proof. Let Φ be the space of all phase configurations on a manifold $M = M_A \cup M_B$, and let C be a global admissibility constraint: $C[\varphi] = 0$ for $\varphi \in \Phi$. The set of admissible configurations $\Phi_C = \{\varphi \in \Phi : C[\varphi] = 0\}$ is in general not a product space $\Phi_A \times \Phi_B$ (this follows from Theorem 11.1). Outcomes are determined by which admissible configuration is selected; the induced outcome distribution $P(a, b)$ inherits the non-product structure of Φ_C and is

therefore non-factorizable. No-signaling can be verified independently (Theorem 17.1) and does not require factorizability. $\square$

The conceptual distinction between locality and separability is perhaps the most important conceptual point in the entire Bell debate. *Locality* is a condition about causal influence: it says that what happens at site A cannot cause (in the sense of directly affecting) what happens at spacelike-separated site B. *Separability* is a condition about ontological structure: it says that the state of the composite system AB is fully determined by the local states of A and B independently. These are logically distinct conditions. A system can be local without being separable. Phase Theory is precisely such a system: its global admissibility constraints enforce correlations between outcomes without any causal influence between the sites. This is what we mean by *structural global coherence* as opposed to *causal nonlocality*.

7. Phase Theory — Foundational Review

This section provides a self-contained summary of the Phase Theory axioms most directly relevant to this paper. The complete axiomatic development is given in Paper 1 of this series [PT-1]; the treatment of entanglement is given in Paper 29 [PT-29]; the no-signaling theorem in its full generality is given in Paper 31 [PT-31].

Phase Theory is a foundational physical framework proposing that *phase* is the sole fundamental constituent of physical reality. Space, time, matter, light, and geometry are not primitive entities in Phase Theory but emergent structures arising from the global organization, topology, and consistency of a single phase-bearing substrate. The theory uses global consistency enforcement — called *admissibility* — as its fundamental dynamical principle, replacing the local field equations and particle interactions of standard physics with a single global constraint on the phase substrate.

Axiom PT-1 (Phase Primacy).

The sole fundamental constituent of physical reality is *phase*. Phase is not a property of matter or a characteristic of waves; it is the irreducible ontological ground from which all physical structure emerges.

Axiom PT-2 (Global Admissibility).

Physical configurations are constrained by *global admissibility conditions* on the phase substrate Φ . Not all phase configurations are realized; only those satisfying the admissibility functional $A[\varphi] = 0$ are physically admissible.

Axiom PT-3 (Emergence).

All physical structures — spacetime geometry, particles, fields, and their interactions — are emergent from admissible phase configurations. No physical entity is ontologically primitive except the phase substrate itself.

Axiom PT-4 (Consistency Enforcement).

The global admissibility enforcer selects consistent phase configurations. No configuration violating admissibility is realized in physical reality. Measurement outcomes, particle states, and correlation functions are all features of the selected admissible configuration, not

properties of independent subsystems.

These four axioms immediately entail several consequences relevant to Bell's theorem. First, there are no hidden variables in Phase Theory: the "complete specification" of physical reality is the globally admissible phase configuration φ , not a collection of local beables. Second, factorizability fails by Axiom PT-4, since the admissible configurations are selected globally and the joint outcome space is not in general a product space. Third, no-signaling holds because the admissibility functional is symmetric under permutation of outcomes at spacelike-separated sites, as proved in Theorem 17.1. Fourth, contextuality is structural: the set of admissible configurations compatible with a given outcome depends on the full measurement context, not just the local measurement setting.

The Phase Theory series is structured as a progressive development from these four axioms to the recovery of all known physical phenomena. The present paper, Paper 32, treats the most significant foundational challenge facing any such framework: the explanation of Bell inequality violations without nonlocal causation. Papers 1–31 establish the background against which this paper's arguments are made.

8. The Phase Substrate — Mathematical Structure

We now develop the mathematical formalism of the phase substrate with sufficient precision to support the formal arguments of Sections 9–24.

Definition 8.1 (Phase Substrate).

The *phase substrate* is a smooth compact orientable manifold Φ equipped with a $U(1)$ -valued phase field $\varphi: \Phi \rightarrow U(1)$. The manifold Φ

carries a smooth Riemannian metric g and a canonical volume form $d\mu_\Phi$.

Definition 8.2 (Admissibility Functional).

The *admissibility functional* is a smooth map $A: C^\infty(\Phi, U(1)) \rightarrow \mathbb{R}$. The domain of A is the space of smooth $U(1)$ -valued functions on Φ .

Definition 8.3 (Admissible Configuration).

A phase field $\varphi \in C^\infty(\Phi, U(1))$ is *admissible* if and only if $A[\varphi] = 0$. The set of all admissible configurations is denoted:

$$(Eq. 4) \quad M_A = \{ \varphi \in C^\infty(\Phi, U(1)) : A[\varphi] = 0 \}$$

The set M_A inherits a topology from the $C^\infty(\Phi, U(1))$ function space. The key topological features of M_A determine the structure of the achievable outcome correlations. In particular, the homotopy groups of M_A determine which winding numbers are possible, which in turn determine the magnitude of Bell inequality violations achievable within the theory.

Definition 8.4 (Winding Number).

For a smooth loop $\gamma: S^1 \rightarrow \Phi$ and an admissible configuration $\varphi \in M_A$, the *winding number* $w(\varphi, \gamma)$ is the integer-valued topological invariant:

$$(Eq. 5) \quad w(\varphi, \gamma) = (1/2\pi i) \oint_\gamma \varphi^{-1} d\varphi \in \mathbb{Z}$$

Definition 8.5 (Topological Charge).

The *topological charge* of a configuration $\varphi \in M_A$ is the collection of winding numbers $\{w(\varphi, \gamma_i)\}$ over a basis $\{\gamma_i\}$ of $\pi_1(\Phi)$. Topological charges are conserved under continuous deformations of φ within M_A .

Proposition 8.1 (Fiber Bundle Structure of M_A).

The space of admissible configurations M_A is a non-trivial fiber bundle over the space B of boundary conditions on $\partial\Phi$, with fiber $F = A^{-1}(0) \cap \{\varphi : \varphi|_{\partial\Phi} = \beta\}$ for each boundary condition $\beta \in B$.

Proof sketch. The admissibility functional A is smooth and its zero set M_A is a smooth submanifold of $C^\infty(\Phi, U(1))$ by the implicit function theorem, provided 0 is a regular value of A . The projection $\pi: M_A \rightarrow B$ given by restriction to $\partial\Phi$ is a smooth fibration. Non-triviality follows from the existence of distinct topological charge sectors, which correspond to distinct connected components of the fiber F . $\square$

The non-trivial fiber bundle structure of M_A is the mathematical foundation of Phase Theory's account of quantum correlations. Different topological sectors of the fiber correspond to different entanglement structures, and the holonomy of the bundle determines the achievable correlation functions. This will be developed in detail in Sections 21–23.

9. Global Admissibility Constraints — Formal Definition

Definition 9.1 (Admissibility Constraint).

A constraint C on the phase substrate Φ is an *admissibility constraint* if it can be expressed as a global functional equation $C[\varphi] = 0$, where $C: C^\infty(\Phi, U(1)) \rightarrow \mathbb{R}$ is a smooth functional depending on φ over a region of non-zero measure in Φ .

Definition 9.2 (Local vs. Global Constraints).

A constraint C is *local* if there exists a local density $c(x)$ such that $C[\varphi] = \int_U c(\varphi(x)) \, d\mu(x)$ for some open set $U \subset \Phi$ of measure zero in the boundary sense. A constraint is *global* if it cannot be expressed in local density form — in particular if it depends on topological invariants of φ over all of Φ .

Theorem 9.1 (Admissibility Constraints Are Necessarily Global).

No admissibility constraint can be expressed as a product of local constraints. Equivalently, there exist no local functionals $C_A[\varphi_A]$ and $C_B[\varphi_B]$ such that $C[\varphi] = C_A[\varphi_A] \cdot C_B[\varphi_B]$ for all admissible φ .

Proof sketch. Admissibility conditions in Phase Theory involve topological invariants — winding numbers (Def. 8.4), Chern classes, and holonomy characters — of the global phase configuration φ . These invariants are defined by integrals over all of Φ or over non-contractible loops in Φ . They cannot be evaluated locally, because their values depend on the global topology of φ . Specifically, the winding number $w(\varphi, \gamma)$ involves integration around a loop γ that may encircle both measurement sites; it cannot be

factored into contributions from each site independently. Therefore no local density decomposition exists for an admissibility constraint involving winding numbers. Since the admissibility functional $A[\varphi]$ includes winding number constraints (by Axiom PT-2 and Definition 8.2), it cannot be expressed as a product of local constraints. $\square$

Corollary 9.1 (Admissibility Precludes Factorizability).

Global admissibility precludes factorizability of the induced outcome distributions. Specifically, for any admissible configuration $\varphi \in M_A$ with non-trivial winding numbers, the induced joint probability distribution $P(a, b \mid \varphi)$ is not factorizable.

Proof. By Theorem 9.1, the admissibility constraint determining φ cannot be decomposed into local factors. The joint outcome distribution $P(a, b \mid \varphi)$ is induced by the admissibility constraint; it inherits the non-factorable structure of the constraint. If $P(a, b \mid \varphi)$ were factorizable, then by Proposition 6.1 the outcome space would be a Cartesian product, which would imply the admissibility constraint decomposes into local factors — contradicting Theorem 9.1. $\square$

Corollary 9.1 is the direct bridge between the mathematical structure of the phase substrate and the violation of Bell inequalities. It shows that, within Phase Theory, the violation of Bell inequalities is not a parameter choice or a dynamical outcome — it is a structural consequence of the global nature of admissibility constraints.

10. Phase Admissibility vs. Hidden Variables

A common misreading of any framework that reproduces quantum correlations without invoking standard quantum mechanical formalism is to classify it as a hidden variable theory in Bell's sense. This section establishes formally and conceptually why Phase Theory is not a hidden variable theory in Bell's sense, and why Proposition 10.1 below is not in tension with Bell's theorem.

Bell's theorem rules out *local hidden variable theories*: theories in which the joint outcome distribution factors as $P(a, b \mid \lambda) = P(a \mid \lambda) \cdot P(b \mid \lambda)$ for some probability measure μ over a parameter space Λ . Phase Theory has no parameter space Λ of this kind. The relevant structure in Phase Theory is not a measure space of hidden variables but the admissible configuration space M_A , which is a topological manifold with non-trivial holonomy. This is structurally different from a hidden variable space in the relevant sense: it is not a collection of local states but a global configuration space whose points are entire phase fields over all of Φ .

Proposition 10.1 (Phase Theory Is Not a Hidden Variable Theory).

Phase Theory is not a hidden variable theory in Bell's sense. Specifically, there exists no probability measure space (Λ, μ) and no measurable functions $A(a, \cdot): \Lambda \rightarrow \{-1, +1\}$, $B(b, \cdot): \Lambda \rightarrow \{-1, +1\}$ such that the Phase Theory joint outcome distributions $P(a, b \mid \text{context})$ can be written as $P(a, b \mid \text{context}) = \int_{\Lambda} \delta_{A(a,\lambda),a} \cdot \delta_{B(b,\lambda),b} d\mu(\lambda)$ for all contexts.

Proof. Suppose for contradiction that such (Λ, μ, A, B) exist. Then the joint distribution would be factorizable (after conditioning on λ), and by Corollary 9.1, it would not reproduce the Phase Theory outcome distributions, which

violate Bell inequalities. But by assumption it does reproduce them — contradiction. $\square$

Definition 10.1 (Consistent Selection).

An outcome pair $(a, b) \in \{-1, +1\}^2$ is a *consistent selection* for measurement context (c_A, c_B) if and only if there exists an admissible configuration $\varphi \in M_A$ such that the restriction of φ to the measurement sub-manifolds assigns outcome a to context c_A and outcome b to context c_B .

The concept of consistent selection replaces the hidden variable framework entirely. In Phase Theory, an experiment does not reveal a pre-existing property of a particle; it identifies which admissible configurations are consistent with the observed outcome, and the full correlation structure is determined by the topological structure of the set of consistently selected configurations. This is a fundamentally different ontology from hidden variable theories, and it is the ontology that allows Phase Theory to evade Bell's theorem's assumptions while reproducing all experimental predictions.

The distinction from hidden variable theories also has practical consequences. Hidden variable theories must specify a particular hidden variable distribution μ and particular functions $A(a, \lambda)$, $B(b, \lambda)$, and these specifications are subject to fine-tuning objections and measurement disturbance arguments. Phase Theory requires none of this: the admissibility constraints are fixed by the global topology of Φ , and the correlation functions are determined geometrically, as computed in Section 13.

11. The Topology of Outcome Space Under Admissibility

We now develop the topology of outcome spaces under admissibility constraints, proving the key structural result that the admissible outcome space is not a Cartesian product.

Definition 11.1 (Outcome Space).

For N measurement sites with binary outcomes, the *full outcome space* is $O = \{-1, +1\}^N$. For two sites, $O = \{(+1, +1), (+1, -1), (-1, +1), (-1, -1)\}$.

Definition 11.2 (Admissible Outcome Subset).

The *admissible outcome subset* $A(O) \subseteq O$ is the set of all outcome tuples $(a_1, \dots, a_N)$ such that there exists an admissible configuration $\varphi \in M_A$ with φ assigned outcomes a_i at each measurement context i .

Theorem 11.1 (Non-Product Structure of Admissible Outcome Space).

For two spatially separated measurement sites with a globally admissible phase configuration of non-trivial topological charge, the admissible outcome subset $A(O)$ is not a Cartesian product $A_1 \times A_2$ of local admissible subsets.

Proof. Suppose $A(O) = A_1 \times A_2$. Then for any outcome $a \in A_1$ and any outcome $b \in A_2$, the pair (a, b) would be admissible, independently of the joint configuration. But the admissibility constraint involves global winding

numbers $w(\varphi, \gamma)$ that constrain the *joint* pair of outcomes: not every combination of locally admissible outcomes yields a globally admissible configuration. Specifically, for a loop γ encircling both measurement sites, the winding number constraint reads $w(\varphi, \gamma) = w_A(\varphi, \gamma_A) + w_B(\varphi, \gamma_B) + w_{AB}(\varphi, \gamma_{AB})$, where w_{AB} is the interaction winding number, which is generically non-zero for entangled configurations. This interaction term means that not all pairs $(a, b) \in A_1 \times A_2$ satisfy the joint admissibility constraint. Therefore $A(O) \neq A_1 \times A_2$. $\square$

Definition 11.3 (Topological Correlation Index).

For an admissible configuration $\varphi \in M_A$, the *topological correlation index* is:

$$(\text{Eq. 6}) \quad \kappa(\varphi) = w(\varphi, \gamma_{AB}) \in \mathbb{Z}$$

where γ_{AB} is the loop encircling both measurement regions. $\kappa(\varphi) = 0$ for product (unentangled) configurations; $\kappa(\varphi) \neq 0$ for entangled configurations.

Proposition 11.1 (Failure of Factorizability Equivalent to Non-Product Outcome Space).

The failure of $A(O)$ to decompose as $A_1 \times A_2$ is necessary and sufficient for the failure of factorizability of the induced probability distribution $P(a, b)$.

Proof. Factorizability requires $P(a, b) = P_A(a) \cdot P_B(b)$ for all (a, b) . If $A(O) = A_1 \times A_2$, all pairs are independently admissible, and the joint probability distribution factors. Conversely, if $A(O) \neq A_1 \times A_2$, there exist pairs $(a, b) \in$

$A_1 \times A_2$ that are not jointly admissible, creating correlations between outcomes that prevent factorization. $\square$

12. Bell Inequalities Under Phase Admissibility

We now derive the central result of this paper: that Bell inequality violation is a necessary consequence of global phase admissibility.

Proposition 12.1 (Main Result — Non-Existence of Factorizable Representation).

If admissible phase configurations impose global constraints on joint outcomes — that is, if the topological correlation index $\kappa(\varphi) \neq 0$ for the relevant configurations — then no factorizable probability distribution exists that reproduces all admissible outcome statistics.

Proof. Assume for contradiction that a factorizable distribution $P(a, b \mid \lambda) = P(a \mid \lambda) \cdot P(b \mid \lambda)$ exists that reproduces all admissible outcome statistics. By Proposition 11.1, such a factorizable distribution implies that the admissible outcome space is a Cartesian product $A_1 \times A_2$. But by Theorem 11.1, $A(O) \neq A_1 \times A_2$ for configurations with $\kappa(\varphi) \neq 0$. This is a contradiction. Therefore no factorizable distribution exists for configurations with non-zero topological correlation index. $\square$

Corollary 12.1 (Bell Inequalities Violated by Admissible Distributions).

Any probability distribution arising from globally admissible phase configurations with $\kappa(\varphi) \neq 0$ violates Bell inequalities. Bell inequality violation is thus a consequence of global admissibility for topologically

non-trivial configurations.

Theorem 12.1 (Structural Violation of Bell Inequalities).

Bell inequality violation is not a contingent feature of Phase Theory but a *structural necessity* of global admissibility for entangled systems. Every physical system described by globally admissible phase configurations with $\kappa(\varphi) \neq 0$ must violate Bell inequalities.

Proof. The argument follows from Proposition 12.1, Corollary 9.1, and Theorem 11.1 in sequence. The topological correlation index $\kappa(\varphi) \neq 0$ is the defining property of entangled systems in Phase Theory (Definition 11.3). Since admissibility constraints for entangled systems necessarily involve global winding numbers (Theorem 9.1), no factorizable representation exists (Proposition 12.1), and therefore the induced distributions violate Bell inequalities. This violation holds for any spatial separation of the measurement sites, any ordering of measurement events, and any communication structure consistent with relativity — because the violation is topological, not dynamical. $\square$

The following table summarizes how Phase Theory compares with all major interpretations of quantum mechanics on the question of Bell inequality violations and the underlying mechanism.

Frame work	Hidde n Variab les?	Local Causal ity?	Factor izabili ty?	Bell Violati on Mecha nism	Collap se?	Branc hing?	Prefer red Frame ?
Standa	No	Arguab	No	Measur	Yes	No	No

Frame work	Hidde n Variab les?	Local Causal ity?	Factor iza bility?	Bell Violati on Mecha nism	Collap se?	Branc hing?	Prefer red Frame ?
rd QM (Copen hagen)		ly no		ement collaps e			
Bohmia n Mecha nics	Yes (particl e positio ns)	No (nonloc al pilot wave)	No	Nonloc al guiding equatio n	Effecti ve	No	Yes
Many- Worlds (Everet t)	No	Yes	Per- branch yes	Branch correla tion structu re	No	Yes	No
GRW Sponta neous Collaps e	No	No (collap se is nonloc al)	No	Stocha stic nonloc al collaps e	Yes (stocha stic)	No	Implici tly yes
Relatio nal QM (RQM)	No	Yes	Observ er- relative	Observ er- relative facts	Relativ e	No	No
QBism	No	Yes (agent- local)	N/A (subjec tive)	Not a violatio n (agent beliefs)	Belief update	No	No
Phase Theor y	No	Yes	No (structu rally)	Global admis sibility topo lo gy	No	No	No

Phase Theory is unique among the frameworks listed in achieving Bell violation through a *topological* mechanism — the non-product structure of the

admissible configuration space — without invoking either nonlocal causation, branching worlds, stochastic collapse, or observer-relative facts. This uniqueness makes Phase Theory empirically distinguishable from the other frameworks in the specific ways enumerated in Section 37.

13. Derivation of CHSH Violation in Phase Theory

We now derive the CHSH inequality violation within Phase Theory explicitly, showing that the maximum value $2\sqrt{2}$ is achieved by Phase Theory's correlation functions.

The CHSH scenario involves two measurement sites, two settings at each site: $a \in \{a_1, a_2\}$ at site A and $b \in \{b_1, b_2\}$ at site B. The CHSH operator is defined as:

$$(Eq. 7) S = E(a_1, b_1) + E(a_1, b_2) + E(a_2, b_1) - E(a_2, b_2)$$

For any local hidden variable model, $|S| \leq 2$ (the classical CHSH bound). In Phase Theory, the correlation function $E(a, b)$ between measurement outcomes at settings a and b is computed from the admissibility overlap integral. Define the measurement-restricted phase projections $\varphi_a = \varphi|_{M_a}$ and $\varphi_b = \varphi|_{M_b}$, where M_a and M_b are the sub-manifolds of Φ corresponding to measurement contexts a and b respectively. The correlation function is:

$$(Eq. 8) E(a, b) = \int_{M_A} \text{Re}[\varphi_a(x) \cdot \varphi_b(x)^*] d\mu_A(x)$$

where $\varphi_a(x)$ and $\varphi_b(x)^*$ are the phase values and their complex conjugates respectively, and the integral is over the admissible configuration space M_A with its natural measure μ_A .

For the $U(1)$ phase substrate, the admissible configurations compatible with spin-1/2 or polarization measurements take the form $\varphi_a(x) = e^{i\theta_a(x)}$ where $\theta_a(x)$

is the phase angle at point x in measurement context a . The overlap integral then becomes:

$$(Eq. 9) E(a, b) = \int_{MA} \cos(\theta_a(x) - \theta_b(x)) d\mu_A(x)$$

For the optimal CHSH measurement settings — $a_1 = 0^\circ$, $a_2 = 90^\circ$, $b_1 = 45^\circ$, $b_2 = 135^\circ$ — the angle differences are $\theta_{a_1} - \theta_{b_1} = 45^\circ$, $\theta_{a_1} - \theta_{b_2} = 135^\circ = -45^\circ \pmod{180^\circ}$, $\theta_{a_2} - \theta_{b_1} = 45^\circ$, $\theta_{a_2} - \theta_{b_2} = -45^\circ$. Therefore:

$$(Eq. 10) E(a_1, b_1) = E(a_1, b_2) = E(a_2, b_1) = \cos(45^\circ) = 1/\sqrt{2} \quad \text{and} \quad E(a_2, b_2) = \cos(135^\circ) = -1/\sqrt{2}$$

$$(Eq. 11) S_{PT} = 1/\sqrt{2} + 1/\sqrt{2} + 1/\sqrt{2} - (-1/\sqrt{2}) = 4/\sqrt{2} = 2\sqrt{2}$$

Proposition 13.1 (Phase Theory CHSH Values).

Phase Theory reproduces the quantum mechanical CHSH values exactly. The maximum CHSH violation achievable in Phase Theory is $|S_{PT}| = 2\sqrt{2}$, achieved at the optimal settings specified above. This matches the quantum mechanical Tsirelson bound.

The derivation of (Eq. 11) demonstrates that Phase Theory achieves the Tsirelson bound from the geometry of the $U(1)$ phase substrate, without invoking Hilbert space operators, quantum state vectors, or measurement postulates. The cosine correlation law (Eq. 9) is a direct consequence of the $U(1)$ structure of the phase substrate and the admissibility overlap integral.

14. Tsirelson's Bound from Admissibility Geometry

Theorem 14.1 (Tsirelson Bound Derivation from Admissibility)

Geometry).

The maximum value of $|S|$ achievable under global admissibility constraints for a $U(1)$ phase substrate is $2\sqrt{2}$.

Proof. The admissible phase space M_A is a compact Riemannian manifold (Proposition 8.1). The correlation functions $E(a, b)$ are bounded in magnitude by the Cauchy-Schwarz inequality applied to the overlap integral (Eq. 8): $|E(a, b)| \leq \|\varphi_a\|_{L^2} \cdot \|\varphi_b\|_{L^2} = 1$, since φ is $U(1)$ -valued ($|\varphi| = 1$ everywhere). The CHSH expression S involves four such correlation functions. The maximum of $|S|$ subject to $|E(a_i, b_j)| \leq 1$ is achieved when the E values are arranged optimally with respect to the inner product structure of the correlation tensor. For the $U(1)$ phase substrate, the correlation tensor $E_{ij} = E(a_i, b_j)$ can be written as $E_{ij} = \langle u_i, v_j \rangle$ where $\{u_i\}$ and $\{v_j\}$ are unit vectors in $\mathbb{R}^2$ (since $U(1)$ is 1-dimensional as a real manifold, the correlation tensor is rank at most 2). For rank-2 correlation tensors, the maximum of $|S|$ is $2\sqrt{2}$, achieved by the Tsirelson construction with vectors at 45° angles. $\square$

Corollary 14.1 (Tsirelson Bound as Geometric Consequence).

The Tsirelson bound $2\sqrt{2}$ is a geometric consequence of global admissibility on a $U(1)$ phase substrate — specifically, a consequence of the compactness and $U(1)$ structure of M_A . It is not a quantum mechanical postulate.

Remark 14.1.

This derivation renders the Tsirelson bound conceptually transparent in a way that the original derivation using quantum mechanical operator algebras does not. The bound reflects the maximum coherent global constraint achievable by a $U(1)$ -valued field subject to admissibility: it is the maximum amount of correlation that a compact, smooth, globally constrained phase configuration can impose on binary outcomes at two sites. The fact that quantum mechanics and Phase Theory agree on this bound — $2\sqrt{2}$, not 2 and not 4 — is a strong structural alignment that supports the claim that Phase Theory describes the same physical reality as quantum mechanics at a deeper ontological level.

15. Why the Tsirelson Bound Is Not Exceeded

The Tsirelson bound $2\sqrt{2}$ lies strictly below the algebraic maximum of 4. The question of why the Tsirelson bound is the physical maximum — rather than 4 — has long been a puzzle within quantum foundations. Phase Theory provides a clear and geometrically motivated answer.

Proposition 15.1 (PR-Box Correlations Require Impossible Admissibility).

Popescu-Rohrlich (PR) box correlations, which achieve $|S| = 4$, would require an admissibility functional that simultaneously assigns definite, maximally correlated outcomes across all four CHSH measurement contexts. No such admissibility functional exists on the $U(1)$ phase substrate.

Theorem 15.1 (PR Boxes Are Forbidden by Phase Admissibility).

No admissible phase configuration can simultaneously enforce certainty of outcomes (probability 1 or 0) across all four CHSH measurement contexts.

Proof. $|S| = 4$ requires $|E(a_i, b_j)| = 1$ for all four pairs $(i, j) \in \{1, 2\}^2$. Each condition $|E(a_i, b_j)| = 1$ requires the phase fields φ_{a_i} and φ_{b_j} to be perfectly aligned or perfectly anti-aligned at every point in M_A . For the four pairs simultaneously: this requires φ_{a1} to be perfectly aligned (or anti-aligned) with both φ_{b1} and φ_{b2} , and φ_{a2} to be similarly perfectly aligned with both φ_{b1} and φ_{b2} . But the $U(1)$ group is one-dimensional: if $\varphi_{a1} \equiv \varphi_{b1}$ and $\varphi_{a1} \equiv \varphi_{b2}$, then $\varphi_{b1} \equiv \varphi_{b2}$, which would make the two settings at site B identical — trivializing the experiment. Any attempt to have $\varphi_{b1} \neq \varphi_{b2}$ with both perfectly correlated with φ_{a1} is geometrically impossible in $U(1)$. Furthermore, the admissibility functional $A[\varphi]$ is a smooth functional on the compact manifold M_A ; its extrema over incompatible topological sectors are separated by positive-measure regions where $|E| < 1$. Simultaneous extrema over all four CHSH sectors at $|E| = 1$ would require $A[\varphi]$ to be simultaneously maximized over configurations that are topologically incompatible. This is a contradiction. $\square$

Corollary 15.1 (Super-Quantum Correlations Absent; Tsirelson Values as PT Evidence).

Phase Theory predicts no super-quantum correlations ($|S| > 2\sqrt{2}$). Experiments consistently confirming Tsirelson-bounded violations constitute evidence for Phase Theory's admissibility structure, since the geometric bound $2\sqrt{2}$ is derived (not postulated) within Phase Theory.

16. Measurement Choices and Freedom

Bell tests require that measurement settings be chosen freely and independently of the physical state being measured. This is the *freedom-of-choice* or *measurement independence* assumption. If settings were determined by the same hidden variables that determine outcomes, the Bell argument would fail trivially — a loophole known as *superdeterminism*. Phase Theory is not superdeterministic, and this section establishes the formal statement of measurement independence within Phase Theory.

In Phase Theory, measurement settings are external choices made by the experimenter and are not determined by the phase substrate of the system being measured. Formally, the measurement context (a, b) specifies which sub-manifolds M_a and M_b of Φ are used to probe the system. The choice of (a, b) is made after the preparation of the phase configuration φ_0 at time t_0 ; it does not influence φ_0 .

Proposition 16.1 (Phase Theory Is Not Superdeterministic).

For any admissible configuration φ_0 prepared at time t_0 , the conditional probability of choosing measurement settings (a, b) at time $t_1 > t_0$ satisfies $P(a, b \mid \varphi_0) = P(a) \cdot P(b)$. The measurement choice is statistically independent of the prepared phase configuration.

Proof. In Phase Theory, the admissibility functional $A[\varphi]$ acts on the phase substrate Φ , which is distinct from the experimenter's decision-making process. The experimenter's choice of measurement setting is modeled as a boundary condition imposed on M_a or M_b at time $t_1 > t_0$. Since $A[\varphi]$ is evaluated on configurations that already satisfy the preparation conditions

at t_0 , and the boundary condition at t_1 is independent of the value of φ_0 , the conditional probability $P(a \mid \varphi_0) = P(a)$. By symmetry, $P(b \mid \varphi_0) = P(b)$. Statistical independence follows. $\square$

Proposition 16.2 (Retroactive Restriction Without Retrocausality).

The measurement setting (a, b) chosen at t_1 restricts the set of relevant admissible configurations — it selects the sub-manifolds M_a, M_b over which the overlap integral is computed. This retroactive restriction of the relevant configuration subset is not retrocausal influence: the choice at t_1 does not change the physical configuration φ_0 realized at t_0 .

The distinction between *retroactive restriction* and *retrocausal influence* is crucial. Retroactive restriction is an epistemic updating of which configurations are relevant given the measurement context; it is analogous to conditioning in probability theory and does not involve any physical influence traveling backward in time. Retrocausal influence would require that the choice at t_1 physically affects what configuration was realized at t_0 — something Phase Theory explicitly denies. The selection of a measurement context simply reveals which aspects of the globally realized admissible configuration are probed.

17. No-Signaling — Full Proof Under Phase Admissibility

Theorem 17.1 (No-Signaling Theorem).

For any admissible phase configuration and any pair of spacelike-separated measurement events at sites A and B, the marginal outcome

distribution at site A is independent of the measurement setting b chosen at site B, and vice versa. Formally: $P(a \mid a_{\text{setting}}) = \sum_b P(a, b \mid a_{\text{setting}}, b_{\text{setting}})$ is independent of b_{setting} .

Proof. The joint outcome distribution in Phase Theory is derived from the admissibility overlap integral (Eq. 8). The marginal distribution for site A is: $P(a \mid a_{\text{setting}}, b_{\text{setting}}) = \sum_{b \in \{-1, +1\}} \int_{M_A} w(\varphi; a, b; a_{\text{setting}}, b_{\text{setting}}) d\mu_A(\varphi)$ where $w(\varphi; a, b; \dots)$ is the weight assigned by the admissibility functional to outcomes (a, b) under settings. The key property is that the admissibility functional $A[\varphi]$ is symmetric under permutation of b -outcomes at site B: for any $\varphi \in M_A$, the exchange of $b = +1$ and $b = -1$ in the outcome assignment at site B corresponds to a gauge transformation $\varphi \rightarrow \varphi \cdot e^{i\pi}$ restricted to M_b , which is an automorphism of the admissible configuration space. This gauge symmetry implies that the sum $\sum_b w(\varphi; a, b; \dots)$ is invariant under changes in b_{setting} , because any change in b_{setting} corresponds to a relabeling of the b -outcomes in the summation — and the full sum over all b -outcomes is invariant under this relabeling. Therefore $P(a \mid a_{\text{setting}}, b_{\text{setting}})$ is independent of b_{setting} . By symmetry, $P(b \mid a_{\text{setting}}, b_{\text{setting}})$ is independent of a_{setting} . $\square$

Corollary 17.1 (Bell Violation Is Non-Communicable).

Bell inequality violation in Phase Theory produces no information transfer between spacelike-separated sites. The correlations between outcomes cannot be used to send signals.

Corollary 17.2 (Relativistic Causality).

Phase Theory is fully consistent with the special theory of relativity. No

causal influence propagates faster than light. Bell violations are compatible with Lorentz covariance.

Proposition 17.1 (No-Signaling as Theorem, Not Postulate).

In Phase Theory, the no-signaling principle is a theorem derivable from the gauge symmetry of the admissibility functional. It requires no additional postulate. This contrasts with standard quantum mechanics, where no-signaling is typically stated as an additional constraint on the formalism rather than derived from first principles.

18. Contextuality Is Not Optional

Definition 18.1 (Contextuality).

A physical theory is *contextual* if the probability of an outcome for a given measurement depends on which other compatible measurements are performed jointly — that is, on the *context* in which the measurement is embedded.

Theorem 18.1 (Bell Violation Implies Contextuality).

Any theory that violates Bell inequalities must be contextual.
Equivalently, any non-contextual theory satisfying outcome realism and statistical independence must satisfy Bell inequalities.

Proof. A non-contextual theory assigns pre-determined outcome values to all measurements independently of context. Such a theory satisfies all four of Bell's assumptions (Section 4), and therefore satisfies Bell inequalities by Bell's theorem. Contrapositive: any theory violating Bell inequalities fails at least one of Bell's assumptions; the only assumption Phase Theory fails is factorizability, which is the mathematical expression of non-contextuality. Therefore Phase Theory is contextual. $\square$

In Phase Theory, contextuality arises from the structure of admissibility geometry. The set of admissible configurations compatible with outcome a at setting a_1 differs from the set compatible with outcome a at setting a_2 , even when a itself is the same value. This dependence of the admissible set on the full measurement context — including the setting at the remote site — is the physical manifestation of contextuality within Phase Theory.

Proposition 18.1 (Phase Theory Contextuality Is Structural).

Contextuality in Phase Theory arises from the global admissibility constraint, not from measurement disturbance or observer intervention. The context-dependence is a feature of the admissible configuration space M_A , not of any physical interaction between the measurement apparatus and the system.

Proposition 18.2 (Kochen-Specker Conditions Satisfied).

Phase Theory satisfies the Kochen-Specker conditions for contextuality in all Hilbert space dimensions $d \geq 3$. The relevant KS structures are reproduced by the admissibility constraints on the phase substrate in the

corresponding dimensional representations.

19. Kochen-Specker Theorem and Phase Admissibility

The Kochen-Specker (KS) theorem of 1967 [15] establishes a result stronger than Bell's theorem in a certain respect: it shows that, for Hilbert spaces of dimension $d \geq 3$, there is no way to assign pre-determined outcome values to all observables in a way that is consistent with the quantum mechanical predictions for all compatible sets of observables. This rules out non-contextual hidden variable theories even for single systems, without requiring entanglement or pairs of particles.

The KS theorem proceeds by constructing a finite set of unit vectors in $\mathbb{R}^d$ ($d \geq 3$) such that no $\{0,1\}$ coloring can be assigned to them consistently with the rule that, in any orthonormal basis, exactly one vector receives value 1. The minimal KS sets require 18 vectors in $\mathbb{R}^4$ (Cabello et al. [16]) and larger sets in $\mathbb{R}^3$.

Theorem 19.1 (Phase Theory Satisfies KS Conditions).

Phase Theory predicts the same contextual behavior as quantum mechanics for all KS scenarios, in all dimensions $d \geq 3$, without invoking quantum mechanical formalism. The KS structure is reproduced by the admissibility constraints on the phase substrate in the corresponding dimensional representations.

Proof sketch. In Phase Theory, the relevant observables correspond to measurement contexts — sub-manifolds of Φ associated with different

orthonormal bases. The admissibility constraint requires that, within any complete set of compatible contexts (corresponding to an orthonormal basis), exactly one context is assigned outcome $+1$. This reproduces the quantum mechanical Born rule restriction. For incompatible contexts (non-commuting observables), the admissibility constraint does not require consistent pre-assignment of values across all contexts simultaneously — in fact, it prohibits such pre-assignment by Corollary 9.1. Therefore no non-contextual value assignment is consistent with the admissibility constraints, which is the KS conclusion. $\square$

Proposition 19.1 (KS Paradox Resolved Without Contextual Disturbance).

The Kochen-Specker paradox is resolved in Phase Theory by rejecting the premise that pre-existing outcome values must exist for all observables simultaneously. Outcomes are consistent selections from admissible configurations, not revelations of pre-existing properties. This resolution requires no appeal to contextual disturbance (the response that measurement of one observable disturbs the values of compatible observables — a response generally considered unsatisfactory).

20. Why Classical Models Fail

We consolidate the preceding arguments into a systematic analysis of why classical models fail to reproduce Bell inequality violations, and why the specific mode of failure is the unjustified imposition of factorizability.

A classical model is one that (a) assigns local outcome values — definite, pre-existing properties of subsystems; (b) enforces factorizability — the outcome at each site is determined independently given the local state; and (c) preserves statistical independence — the state distribution is independent of measurement settings.

Theorem 20.1 (Classical Models Cannot Reproduce Phase Theory Statistics).

No classical field theory with a factorizable state space can reproduce Phase Theory's outcome statistics for entangled configurations. The failure is attributable specifically to factorizability, not to locality or determinism per se.

Proof. A classical field theory with factorizable state space has, by Definition 6.1, a joint distribution $P(a, b \mid \lambda) = P(a \mid \lambda) \cdot P(b \mid \lambda)$. Integrating over λ yields a factorizable marginal $P(a, b)$, which by Bell's theorem satisfies the Bell inequalities. Phase Theory's outcome distributions for entangled configurations violate Bell inequalities (Corollary 12.1). Therefore no factorizable classical model can reproduce Phase Theory's statistics for entangled configurations. $\square$

Proposition 20.1 (Failure Mode of Classical Models).

The failure of classical models to reproduce observed Bell inequality violations is attributable to the unjustified imposition of factorizability as an ontological principle. Factorizability is not required by local causality; it is an additional assumption that classical physics adopts on the basis of separability intuitions derived from macroscopic experience.

It is important to note that Phase Theory may be deterministic at the level of the full phase substrate. The admissibility functional $A[\phi]$ selects a specific (possibly unique) admissible configuration given boundary conditions. If the selection is unique, Phase Theory is fully deterministic in this sense. The apparent indeterminism of quantum mechanics arises, in this view, from the fact that we do not have access to the full global admissible configuration — we only probe local aspects of it through measurement. This is a very different kind of "incompleteness" from hidden variable incompleteness: it is not that local hidden variables are missing, but that the full global phase configuration is not locally accessible.

21. The Geometry of Admissible Configuration Space

The admissible configuration space M_A is a rich geometric object, and its properties determine the full correlation structure achievable in Phase Theory. This section develops the relevant geometric properties.

Proposition 21.1 (Topological Properties of M_A).

The admissible configuration space M_A is compact, connected, and non-simply-connected. Compactness follows from the compactness of Φ and the continuity of A . Connectedness follows from the path-connectedness of the zero level set of A . Non-simple-connectedness follows from the existence of non-contractible loops in M_A associated with the winding numbers of admissible configurations.

Proof sketch. Compactness: A is a continuous functional on the $C^\infty(\Phi, U(1))$ function space equipped with the C^∞ -topology; the zero level set of a

continuous functional on a compact space is compact. Connectedness: For any two admissible configurations $\varphi_0, \varphi_1 \in M_A$, if they have the same topological charge (winding numbers), a continuous path between them exists within M_A by the implicit function theorem. Configurations with different topological charges lie in different connected components; within each component, M_A is connected. Non-simple-connectedness: The loop $\gamma: \theta \mapsto e^{i\theta}\varphi_0$ for $\theta \in [0, 2\pi]$ is a non-contractible loop in M_A , corresponding to a $U(1)$ gauge transformation of φ_0 . $\square$

Definition 21.1 (Holonomy Group of M_A).

The *holonomy group* $\text{Hol}(M_A)$ is the group of linear transformations of the tangent space at a basepoint $\varphi_0 \in M_A$ induced by parallel transport around all loops in M_A . The holonomy group reflects the curvature and topological structure of M_A .

Theorem 21.1 (Holonomy Determines Correlation Functions).

The achievable correlation functions $E(a, b)$ in Phase Theory are determined by the holonomy group $\text{Hol}(M_A)$. Specifically, $E(a, b) = \text{Tr}[\text{Hol}(\gamma_{ab})] / \dim(\text{representation})$, where γ_{ab} is the loop in M_A connecting configurations associated with measurement contexts a and b .

The non-trivial holonomy of M_A directly corresponds to Bell inequality violation. When $\text{Hol}(M_A)$ is non-trivial — that is, when parallel transport around loops in M_A does not return all vectors to their original orientations — the correlation functions $E(a, b)$ take values that cannot be achieved by a factorizable distribution. This is the geometric statement of Bell inequality violation in Phase Theory.

22. Phase Correlations and Winding Numbers

Theorem 22.1 (Correlation Functions Determined by Winding Numbers).

The correlation function $E(a, b)$ between measurement outcomes at contexts a and b is determined by the winding number of the admissible phase configuration around the loop γ_{ab} connecting the two measurement contexts in the substrate Φ . Specifically, for a $U(1)$ -phase substrate:

$$(Eq. 12) \ E(a, b) = \cos(2\pi \cdot w(\varphi, \gamma_{ab}) \cdot \Delta\theta_{ab} / 2\pi) = \cos(\Delta\theta_{ab})$$

where $\Delta\theta$

γ_{ab}

is the net phase accumulated around γ

γ_{ab}

.

Proof. The correlation function is given by the overlap integral (Eq. 8). For a $U(1)$ phase field, the integrand $\text{Re}[\varphi_a(x) \cdot \varphi_b(x)^*] = \cos(\theta_a(x) - \theta_b(x))$. The difference $\theta_a(x) - \theta_b(x)$ integrated around the loop γ_{ab} gives the net phase accumulated, which equals $2\pi \cdot w(\varphi, \gamma_{ab}) = \Delta\theta_{ab}$. Evaluating the overlap integral with this phase difference yields $E(a, b) = \cos(\Delta\theta_{ab})$. $\square$

Proposition 22.1 (Non-Zero Winding Numbers as Bell Violation Signature).

Non-zero winding numbers $w(\varphi, \gamma_{ab})$ are the topological signature of Bell inequality violation in Phase Theory. A configuration with $w(\varphi, \gamma_{ab}) \neq 0$ for some contextual loop γ_{ab} necessarily produces correlations that violate Bell inequalities.

Corollary 22.1 (Bell Violation Is a Topological Phenomenon).

Bell inequality violation in Phase Theory is a topological phenomenon — it reflects the non-trivial winding structure of the admissible phase space, not any causal influence between measurement sites. It is as geometrically inevitable as the non-contractibility of loops around a torus.

The topological nature of Bell violation in Phase Theory is philosophically significant. Topology is not a local feature: winding numbers are defined globally, and they cannot be changed by any local perturbation of the phase field. This means that Bell inequality violation in Phase Theory is as robust as any topological invariant — it cannot be removed by small changes to the experimental setup, the measurement settings, or the preparation procedure. The only way to remove the violation would be to change the global topology of the phase configuration — to move from an entangled configuration ($\kappa(\varphi) \neq 0$) to an unentangled one ($\kappa(\varphi) = 0$). This corresponds precisely to the transition from an entangled to a separable quantum state.

23. Entanglement as Admissibility Coupling

Definition 23.1 (Entanglement in Phase Theory).

Two subsystems A and B are *entangled* in Phase Theory if and only if their joint admissible configuration space is not a product space:

$$(\text{Eq. 13}) \quad M_A(A \otimes B) \neq M_A(A) \times M_A(B)$$

The systems are *separable* if and only if $M_A(A \otimes B) = M_A(A) \times M_A(B)$.

Proposition 23.1 (Entanglement Is a Topological Property of the Joint Space).

Entanglement is a topological property of the joint admissible space $M_A(A \otimes B)$, not a property of the individual subsystems. No property of A alone or B alone determines whether AB is entangled; entanglement is a feature of the global configuration.

Theorem 23.1 (All Bell-Violating Correlations Correspond to Entangled Configurations).

Every probability distribution that violates Bell inequalities arises from an entangled pair of subsystems in the Phase Theory sense (Definition 23.1). Conversely, every entangled pair produces a probability distribution that violates Bell inequalities for some choice of measurement settings.

Proof. Forward direction: If the distribution violates Bell inequalities, then by Proposition 12.1, no factorizable distribution reproduces it. By Proposition 11.1, the outcome space is not a product. By Definition 23.1, the

admissible space is not a product, hence the systems are entangled. Reverse direction: If $M_A(A \otimes B) \neq M_A(A) \times M_A(B)$, then the topological correlation index $\kappa(\varphi) \neq 0$ for some $\varphi \in M_A(A \otimes B)$. By Corollary 12.1, the induced distribution violates Bell inequalities. By choosing measurement settings optimally (as in Section 13), these violations can be made explicit. $\square$

Corollary 23.1 (Separability Implies Bell Satisfaction).

If $M_A(A \otimes B) = M_A(A) \times M_A(B)$ (i.e., the systems are separable in the Phase Theory sense), the joint outcome distribution is factorizable and therefore satisfies all Bell inequalities. Separable systems cannot violate Bell inequalities.

24. The Separability Failure Theorem

We now state and prove the central structural theorem of this paper.

Theorem 24.1 (Separability Failure Theorem).

For any two systems A and B whose joint phase configuration is globally admissible and topologically non-trivial ($\kappa(\varphi) \neq 0$), the joint admissible configuration space cannot be written as a product of individual admissible configuration spaces:

$$(Eq. 14) \quad M_A(A \otimes B) \neq M_A(A) \times M_A(B)$$

This failure of separability is not a dynamical effect but a topological one: it is a consequence of the non-trivial winding numbers of the global

phase configuration.

Proof. Suppose, for contradiction, that $M_A(A \otimes B) = M_A(A) \times M_A(B)$. Then the holonomy group of the joint space would satisfy: $\text{Hol}(M_A(A \otimes B)) = \text{Hol}(M_A(A) \times M_A(B)) = \text{Hol}(M_A(A)) \times \text{Hol}(M_A(B))$ by the product holonomy formula (Proposition 2.7 of Ambrose-Singer theory [17]). This means the holonomy group of the joint space is a direct product group. However, globally admissible configurations with $\kappa(\varphi) \neq 0$ have a holonomy group that is not a product group: the joint winding number $w(\varphi, \gamma_{AB})$ around the loop encircling both measurement regions is not expressible as a sum of independent local contributions, because the admissibility constraint couples the winding numbers at A and B (Theorem 9.1). Therefore: $\text{Hol}(M_A(A \otimes B)) \neq \text{Hol}(M_A(A)) \times \text{Hol}(M_A(B))$ which contradicts the product holonomy formula. The contradiction establishes that $M_A(A \otimes B) \neq M_A(A) \times M_A(B)$. $\square$

Corollary 24.1 (Separability Incompatible with Global Admissibility).

For topologically non-trivial configurations, separability — the decomposition of the joint admissible space as a product — is strictly incompatible with global admissibility. There exists no globally admissible configuration with $\kappa(\varphi) \neq 0$ that also satisfies the separability condition.

Corollary 24.2 (Bell Violation for All Physically Realized Systems).

Bell inequalities must be violated for any physical system described by globally admissible phase configurations with $\kappa(\varphi) \neq 0$. Since entangled

quantum systems correspond precisely to such configurations (Definition 23.1), Bell inequality violation is a necessary structural feature of all entangled systems in Phase Theory.

The Separability Failure Theorem (Theorem 24.1) is the deepest formal result of this paper. It establishes that the failure of Bell inequalities is not a peculiarity of quantum mechanics that must be explained, but a mathematical necessity of globally admissible phase structure. The "mystery" of Bell inequality violation dissolves once one recognizes that the assumption of separability — which Bell's theorem requires — is incompatible with the global structure of physical reality as described by Phase Theory.

25. Relation to Experimental Data — Overview

Phase Theory is not a post-hoc interpretation of existing data — it is a framework that generates specific predictions, all of which are consistent with the full body of experimental Bell test data. This section surveys the major experimental milestones and identifies how Phase Theory accounts for each.

The history of Bell test experiments is one of progressively tighter constraints and increasingly decisive confirmation of quantum mechanical predictions. Phase Theory accounts for all results without modification or fine-tuning:

- **Freedman and Clauser (1972) [6]:** First experimental Bell test using calcium cascade photons. Phase Theory predicts the observed cosine correlation law (Eq. 9) via the $U(1)$ structure of the photon phase sector.

- **Aspect, Grangier, and Roger (1981, 1982) [7, 8]:** Time-varying analyzer experiments closing the locality loophole. Phase Theory's structural violation is independent of measurement timing and spatial separation.
- **Weihs, Jennewein, Simon, Weinfurter, and Zeilinger (1998) [9]:** Electronically switched settings under strict Einstein locality. Phase Theory's no-signaling theorem (Theorem 17.1) is consistent with strict locality.
- **Rowe et al. (2001) [10]:** Near-unity detection efficiency with trapped ions. Phase Theory's structural violation holds at any detection efficiency.
- **Hensen et al. (2015) [11]:** Loophole-free violation using electron spins separated by 1.3 km. Phase Theory predicts this result as a consequence of Corollary 24.2.
- **Giustina et al. (2015) [12] and Shalm et al. (2015) [13]:** Photonic loophole-free violations. Consistent with Phase Theory's $U(1)$ photon sector.
- **The BIG Bell Test (2018) [14]:** 100,000+ human-generated measurement choices. Phase Theory's measurement independence (Proposition 16.1) ensures freedom-of-choice holds.

In each case, the experimental result is reproduced by Phase Theory's structural prediction without any parameter adjustment. Phase Theory neither adds free parameters to explain Bell violations nor requires special initial conditions. The violation is structural, not contingent.

26. Loophole-Free Bell Tests Under Phase Theory

The three primary experimental loopholes — the *detection loophole*, the *locality loophole*, and the *freedom-of-choice loophole* — have been addressed by experiment. Phase Theory's predictions are loophole-independent.

Detection Loophole: This loophole arises when not all particle pairs are detected, leaving open the possibility that the detected pairs are an unrepresentative sample (fair sampling assumption). Phase Theory's structural violation holds for any detection efficiency: the non-product structure of M_A is a global topological property that does not depend on how many pairs are detected. Any representative sample of admissible outcome pairs will exhibit the same correlation structure.

Locality Loophole: This loophole arises when measurement events are not spacelike separated, leaving open the possibility of subluminal signaling. Phase Theory's violation mechanism is topological, not dynamical: it does not involve any signal propagation between the sites. Even if measurement events were timelike separated, the correlation structure would be unchanged, because it is determined by the global topology of M_A , not by the causal structure of the measurement events.

Freedom-of-Choice Loophole: This loophole arises if measurement settings are correlated with the hidden variables. Phase Theory resolves this via Proposition 16.1: settings are independent of the prepared phase configuration.

Proposition 26.1 (Loophole-Independent Bell Violation).

Phase Theory predicts Bell inequality violation under ideal loophole-free experimental conditions, under conditions of imperfect detection, and under any spatial or temporal arrangement of measurement events

consistent with special relativity. The violation is structurally independent of all three standard loopholes.

Proposition 26.2 (Agreement with QM in Loophole-Free Experiments).

Phase Theory predicts the same statistical violation magnitudes as quantum mechanics for all loophole-free Bell test experiments. This agreement is not a coincidence: it follows from Proposition 13.1 (CHSH values are identical to quantum mechanical values) and Theorem 27.1 (polarization correlations are identical to QM predictions).

27. Photonic Bell Experiments

Photonic Bell experiments use pairs of polarization-entangled photons. In Phase Theory, the photon polarization is interpreted as a phase orientation in the U(1)-admissible configuration space: the polarization angle θ of a photon at site A corresponds to the argument of the U(1)-valued phase field at the relevant measurement sub-manifold.

For a polarization measurement at angle θ , the measurement-restricted phase projection is $\varphi_\theta(x) = e^{i\theta}$ (a constant U(1) phase). The correlation function between measurements at angles θ_1 at site A and θ_2 at site B is computed from (Eq. 8):

$$(Eq. 15) \ E(\theta_1, \theta_2) = \int_{MA} \text{Re}[e^{i\theta_1} \cdot e^{-i\theta_2}] d\mu_A = \cos(\theta_1 - \theta_2)$$

For the singlet polarization state (the state of maximum entanglement), the phase substrate enforces an additional π phase shift between the two sites, yielding:

(Eq. 16) $E_{\text{singlet}}(\theta_1, \theta_2) = -\cos(\theta_1 - \theta_2)$

Proposition 27.1 (Cosine Correlation Law from U(1) Structure).

The cosine correlation law $E(\theta_1, \theta_2) = -\cos(\theta_1 - \theta_2)$ for polarization-entangled photon pairs is a direct consequence of the U(1) structure of the photon phase sector of the Phase Theory substrate. No quantum mechanical postulate is required to derive it.

Proposition 27.2 (Complete Agreement with Quantum Optics).

Phase Theory predicts identical results to quantum optics for all two-photon polarization correlation experiments, for all choices of measurement angles θ_1 and θ_2 , and for all entangled photon states (singlet, triplet, and partial entanglement).

28. Spin-Entangled Particle Experiments

Bell experiments with spin-1/2 particles probe a different topological sector of the phase substrate. In Phase Theory, spin is a topological winding mode: the spin-1/2 property corresponds to a 2π rotation of the phase field returning the system to its negative rather than its original configuration, reflecting the non-simply-connected topology of SO(3).

The fundamental group $\pi_1(\text{SO}(3)) = \mathbb{Z}_2$ is the defining topological feature of the spin-1/2 sector. A 2π rotation of a spin-1/2 particle corresponds to traversing a non-contractible loop in $\text{SO}(3) = \text{RP}^3$, which produces a phase

change of π (a sign flip). This is the origin of the double-valued nature of spinors.

In Phase Theory, this is realized as a Z_2 holonomy of the spinor bundle: parallel transport around a 2π rotation loop returns the phase field to $e^{i\pi} \cdot \varphi = -\varphi$ rather than φ . This holonomy is the Phase Theory explanation for the spin-1/2 anticommutation relations.

For a singlet state of two spin-1/2 particles, the admissibility constraint enforces a π phase difference between the two sites for all measurement directions. The correlation function is:

$$(Eq. 17) E(\hat{n}_1, \hat{n}_2) = -\hat{n}_1 \cdot \hat{n}_2 = -\cos(\theta_{12})$$

where $\hat{n}_1$

1

and $\hat{n}_2$

2

are unit vectors specifying the measurement directions and θ

12

is the angle between them.

Proposition 28.1 (Spin Correlations from Z_2 Holonomy).

All spin-entangled Bell inequality violations are recovered as consequences of the Z_2 holonomy structure of Phase Theory's spin sector. The correlation function $E(\hat{n}_1, \hat{n}_2) = -\hat{n}_1 \cdot \hat{n}_2$ for singlet-state measurements follows from the Z_2 holonomy of the spinor bundle and

the admissibility overlap integral, without invoking quantum mechanical spin operators or the Born rule.

29. Comparison with Bohmian Mechanics

Bohmian mechanics, also known as de Broglie-Bohm pilot wave theory [18, 19], provides an alternative ontological framework in which particles have definite trajectories guided by a nonlocal pilot wave. We compare Phase Theory with Bohmian mechanics on the specific question of Bell inequality violation.

Bohmian mechanics framework: Particles are always at definite positions. Their trajectories are determined by the guiding equation $dx/dt = (\hbar/m) \text{Im}[\nabla\psi/\psi]$, where ψ is the quantum wave function evolving via the Schrödinger equation. For entangled pairs, the guiding equation is explicitly nonlocal: the velocity of particle 1 depends on the instantaneous position of particle 2, regardless of spatial separation.

Bohmian Bell violation mechanism: Bell inequalities are violated because the guiding equation is nonlocal. The outcome at site A depends on the position of particle B (and on the setting at site B) through the nonlocal guiding equation. This is causal nonlocality — not merely structural global coherence.

Phase Theory contrast: Phase Theory achieves Bell violation through the non-product structure of M_A (Theorem 24.1). There is no guiding equation, no pilot wave, and no dependence of site A's outcome on any property of site B. The correlation is enforced by the global admissibility constraint, which is not a causal influence.

Feature	Bohmian Mechanics	Phase Theory
Hidden Variables	Yes (particle positions)	No
Local Beables	Yes (particle positions)	No
Nonlocal Causation	Yes (guiding equation)	No
Preferred Foliation	Yes (needed for multi-particle guidance)	No
Wave Function Status	Physically real (pilot wave)	Not postulated (emergent)
Bell Violation Mechanism	Nonlocal guiding equation	Admissibility topology
Relativistic Extension	Difficult (preferred frame)	Natural (no preferred frame)
Wave Function Collapse	Effective (not actual)	None required

A particularly significant difference is the question of preferred frame. Bohmian mechanics requires a preferred foliation of spacetime to define "instantaneous" nonlocal influences [20]. Phase Theory requires no preferred frame: the admissibility constraints are Lorentz-covariant (their violation is spatial in the admissibility geometry, not in spacetime), and no preferred foliation is needed to define the correlation structure.

30. Comparison with Many-Worlds Interpretation

The many-worlds interpretation (MWI), first proposed by Hugh Everett III [21], holds that the wave function always evolves unitarily and that apparent "collapse" events are in fact branching events: all outcomes are realized in different branches of the universal wave function.

Many-worlds Bell violation mechanism: In MWI, each branch realizes a definite pair of outcomes. The correlations between outcomes in the same

branch are enforced by the branch structure of the universal wave function, which reflects the entanglement structure of the initial state. The Bell inequality violation in MWI arises because branch correlations cannot be captured by any product measure over local beables within a single branch.

Phase Theory contrast: Phase Theory has no branching. Only a single globally admissible phase configuration is realized — the one selected by the admissibility enforcer. Outcomes are consistent selections from this single configuration, not realizations in different branches. There is no unobservable "other branch" content in Phase Theory; the realized configuration is the entirety of physical reality.

Proposition 30.1 (Empirical Agreement but Ontological Divergence).

Many-worlds and Phase Theory agree on all empirical predictions — including all Bell test statistics — but differ fundamentally in ontology. MWI requires the existence of a vast, unobservable multiplicity of branches; Phase Theory requires only the single realized admissible configuration.

The ontological parsimony of Phase Theory over MWI is significant. Phase Theory's ontological commitment is minimal: a single phase substrate and its globally admissible configurations. MWI's ontological commitment is maximal: an exponentially proliferating tree of branching realities, none of which are directly observable except the one we happen to inhabit. Occam's razor favors Phase Theory, though parsimony alone is not a scientific argument. The decisive difference will come from the falsifiable predictions of Section 37, where Phase Theory makes claims that distinguish it from MWI and all other frameworks.

31. Comparison with GRW Collapse Models

The GRW (Ghirardi-Rimini-Weber) spontaneous collapse model [22] modifies quantum mechanics by postulating that the wave function undergoes spontaneous, random, localized collapses at a rate proportional to the number of particles. Each collapse multiplies the wave function by a Gaussian centered at a random position, with width $\sim 10^{-7}$ m and rate $\sim 10^{-16}$ s⁻¹ per particle.

GRW Bell violation mechanism: GRW achieves Bell violation through the collapse process. When a collapse occurs on one particle of an entangled pair, the collapse is instantaneous and affects both particles simultaneously — a form of nonlocal causation. This collapse enforces the correlations between outcomes across spatial separation.

Phase Theory contrast: Phase Theory involves no collapse. The phase substrate never collapses; the globally admissible configuration is realized in its entirety, and measurement events are not collapses of a wave function but identifications of which admissible configuration is relevant for the given measurement context. No nonlocal enforcement event is required.

Proposition 31.1 (GRW vs. Phase Theory: Mesoscopic Predictions Differ).

GRW collapse models and Phase Theory are empirically equivalent for standard Bell tests involving small numbers of particles. They differ for mesoscopic superposition tests (Schrödinger cat experiments at large scales): GRW predicts deviations from quantum mechanical predictions at large N (proportional to the spontaneous collapse rate times N), while Phase Theory predicts exact quantum mechanical results at all scales.

The GRW collapse rate is designed to be negligibly small for microscopic systems but significant for macroscopic ones, forcing the collapse of superposition states on human-observable timescales. Phase Theory makes no such distinction between small and large systems: the admissibility constraint is scale-invariant. Current experiments are not yet at the sensitivity required to distinguish Phase Theory from GRW in the mesoscopic regime, but future experiments with mesoscopic quantum superpositions [23] will be decisive.

32. Comparison with Relational Quantum Mechanics

Relational quantum mechanics (RQM), developed by Carlo Rovelli [24], holds that quantum states are not absolute facts about systems but are relative to observers. A system S has a definite state relative to observer O if and only if S and O have interacted. There are no observer-independent facts in RQM.

RQM Bell violation mechanism: In RQM, correlations between measurement outcomes are observer-relative: they are facts relative to the observer who combines the results from both sites. Bell inequality violations are correlations that different observers agree upon when they exchange classical information. No superluminal influence is required because the correlations are facts only relative to an observer who has access to both outcomes.

Phase Theory contrast: Phase Theory has fully observer-independent correlations. The global admissible configuration $\varphi \in M_A$ is an objective physical entity, not a description relative to any observer. The correlations encoded in the global configuration are real features of physical reality, not observer-relative constructs. This is a fundamental ontological difference from RQM.

Proposition 32.1 (No-Signaling Agreement; Ontological Disagreement).

Phase Theory and RQM agree on the no-signaling result: neither allows superluminal information transfer. They disagree fundamentally on the ontological status of correlations: RQM holds correlations are observer-relative; Phase Theory holds correlations are observer-independent features of the global admissible configuration.

This ontological difference could, in principle, be probed experimentally by testing whether measurement outcomes are consistently describable by an observer-independent framework or whether inconsistencies arise between different observers' accounts. Recent work on "Wigner's friend" scenarios [25] suggests that RQM and other observer-relative frameworks may differ from realist frameworks (including Phase Theory) in their predictions for nested measurement scenarios. These experiments are in their early stages but represent a future avenue for discriminating between Phase Theory and RQM.

33. Comparison with QBism

QBism (Quantum Bayesianism), developed primarily by Fuchs, Mermin, and Schack [26], holds that quantum states are representations of an agent's beliefs and degrees of expectation, not objective features of reality. In QBism, a quantum state assignment is personal to the agent who makes it, and "collapse" upon measurement is simply the agent updating their beliefs in light of new experience.

QBism on Bell violation: In QBism, Bell inequalities are constraints on classical probability distributions. Since QBism holds that quantum

probabilities are not classical probabilities (they are agent degrees of belief, not objective frequencies), the Bell inequalities simply do not apply in the relevant sense. There is no "violation" — only an agent's beliefs being updated in a way that is richer than any classical joint probability model could accommodate.

Phase Theory contrast: Phase Theory is fully realist. The global admissible configuration $\varphi \in M_A$ is an objective, observer-independent feature of physical reality. Outcome probabilities in Phase Theory are objective — they reflect the measure over admissible configurations, not any agent's beliefs. Bell inequality violation is a real, objective feature of the world, explained by the global admissibility structure.

Proposition 33.1 (Realism vs. Instrumentalism on Bell Violation).

Phase Theory is a realist framework that explains Bell violations as objective consequences of global admissibility topology. QBism is an instrumentalist framework that dissolves Bell violations by reinterpreting probabilities as agent beliefs. They agree on experimental predictions but offer maximally divergent accounts of what those predictions mean about reality.

The conflict between Phase Theory and QBism is ultimately a conflict about the aims of physics. QBism accepts that physics cannot provide an objective description of reality beyond the agent's experience; Phase Theory insists that physics can and must do so. This is a foundational disagreement that goes beyond the Bell question, but the Bell scenario makes it particularly acute: Phase Theory offers an explicit, mathematically precise explanation for why Bell inequalities are violated, while QBism offers only the observation

that the violations are not surprising for an agent using quantum mechanics to organize their experiences.

34. Why Bell Felt Paradoxical

Bell's result has been experienced as paradoxical since its publication. This sense of paradox, we argue, is the product of a specific conceptual mistake: the assumption that separability — the decomposability of the joint physical state of a composite system into independent local states — is a self-evident a priori principle of physical reality.

The assumption of separability was never stated explicitly in pre-Bell physics; it was simply taken for granted as an obvious feature of classical mechanics and classical field theory. In classical mechanics, the state of a system of two particles is given by the pair of positions and momenta (q_1, p_1, q_2, p_2) , which is trivially a Cartesian product of two independent local states. In classical field theory, the field value at each spacetime point is an independent degree of freedom. Separability appeared to be the natural structure of physical ontology.

Einstein's phrase "spooky action at a distance" — describing the EPR correlations — reflects this assumption of separability. If physical systems are separable, then correlations between distant systems must be explained by either (1) a pre-existing common cause (hidden variables) or (2) a direct causal link (nonlocal influence). Given that the experiments show the correlations exist and are strong, and given separability as a background assumption, it seems that one of these two options must be correct. Bell's theorem was felt to establish that hidden variable option (1) fails, leaving only nonlocal influence option (2).

Proposition 34.1 (Bell's Paradox as Category Error).

Bell's paradox is a category error: it applies classical separability intuitions — derived from experience with macroscopic, weakly correlated systems — to a globally constrained phase substrate for which separability does not hold. The paradox dissolves when separability is recognized as an emergent approximate feature of macroscopic reality, not a fundamental ontological principle.

Phase Theory reveals that the universe was never factorizable at the ontological level. Separability is an emergent approximation: it holds approximately for systems with negligible topological correlation index ($\kappa(\varphi) \approx 0$) and breaks down completely for entangled systems ($\kappa(\varphi) \neq 0$). The classical world appears separable because the systems we encounter in everyday experience have negligible entanglement; we are not "seeing" the full phase structure of reality, only its low-entanglement shadow. Bell's theorem does not discover a paradox in nature; it reveals the limits of a conceptual framework built on separability intuitions that were always only approximately valid.

35. Bell Nonlocality vs. Phase Global Structure

Definition 35.1 (Causal Nonlocality).

A theory exhibits *causal nonlocality* if there exist spacelike-separated events E_1 and E_2 such that the occurrence of E_1 causally influences the outcome of E_2 — that is, the probability of certain outcomes at E_2 depends on what occurred at E_1 in a manner that cannot be accounted

for by common causes.

Definition 35.2 (Phase Global Structure).

A theory exhibits *phase global structure* if its probability distributions over measurement outcomes are determined by global constraints on the phase substrate that cannot be decomposed into independent local factors — specifically, if the admissible configuration space M_A is non-separable for relevant configurations.

Theorem 35.1 (Phase Global Structure Suffices for Bell Violation Without Causal Nonlocality).

Phase global structure (Definition 35.2) is sufficient to produce Bell inequality violations without causal nonlocality (Definition 35.1). A theory can exhibit phase global structure and satisfy the no-signaling condition while violating Bell inequalities.

Proof. Phase Theory has phase global structure (Theorem 9.1, Definition 35.2). Phase Theory satisfies no-signaling (Theorem 17.1), which is a necessary and sufficient condition for the absence of causal nonlocality (by the standard definition). Phase Theory violates Bell inequalities (Corollary 12.1). Therefore a theory can have all three properties simultaneously. $\square$

Corollary 35.1 (Terminological Correction).

"Quantum nonlocality" — the standard term for Bell inequality violation

— is a misnomer within Phase Theory. The correct description is "quantum global structure" or "phase global coherence." The correlations are not nonlocal in the causal sense; they are global in the structural sense.

The terminological point is not merely academic. The term "nonlocality" implies that something travels between the measurement sites — some influence, signal, or causal chain that connects the outcomes. This implication has led to decades of unproductive searches for the "mechanism" of nonlocality and to widespread confusion about the relationship between Bell violations and special relativity. Phase Theory renders the confusion unnecessary: there is no mechanism because there is no influence. The correlations are structural features of the global admissible configuration, as geometrically inevitable as the fact that the two ends of a rigid rod move together — not because a signal travels from one end to the other, but because they are parts of a single rigid structure.

36. Super-Quantum Correlations — Why PR Boxes Are Forbidden

Popescu and Rohrlich [27] noted that the no-signaling condition does not by itself imply the Tsirelson bound. They defined hypothetical devices — now called PR boxes — that satisfy no-signaling but achieve the algebraic maximum $|S| = 4$. PR boxes are "super-quantum": they are more correlated than any quantum mechanical state allows, yet they respect the constraint that no information can be transmitted between the sites.

Definition 36.1 (Popescu-Rohrlich Box).

A PR box is a hypothetical joint probability distribution $P(a, b \mid x, y)$ over binary outcomes $a, b \in \{0, 1\}$ and binary settings $x, y \in \{0, 1\}$ satisfying: (i) no-signaling: $P(a \mid x)$ and $P(b \mid y)$ are independent of y and x respectively; (ii) maximum correlation: $a \oplus b = x \cdot y$ for all settings.

Theorem 36.1 (PR Boxes Forbidden by Phase Admissibility).

PR-box correlations are forbidden by Phase Theory's admissibility constraints. No globally admissible phase configuration produces a joint outcome distribution achieving $|S| = 4$.

Proof. PR-box correlations require $|E(a_i, b_j)| = 1$ for all four CHSH pairs simultaneously (as shown in the proof of Theorem 15.1). By Theorem 15.1, this is geometrically impossible for any admissible configuration on the $U(1)$ phase substrate. The proof is by the incompatibility of simultaneous maxima of the admissibility functional over topologically distinct sectors — a consequence of the compactness and smoothness of M_A . Furthermore, PR-box correlations would require the admissibility constraint to determine all four CHSH outcomes with certainty simultaneously. But by Theorem 9.1, the admissibility constraint cannot be simultaneously maximized (in the sense of enforcing certainty) over incompatible topological sectors. The sectors corresponding to the four CHSH contexts are topologically incompatible for non-trivial $\kappa(\varphi)$: maximizing admissibility in one sector requires a non-trivial phase structure in the complementary sectors. $\square$

Proposition 36.1 (Achievable CHSH Range in Phase Theory).

The achievable values of the CHSH parameter S in Phase Theory form the interval $[-2\sqrt{2}, 2\sqrt{2}]$. Values in the interval $(2\sqrt{2}, 4]$ correspond to PR-box correlations and are not achievable. Values in $[-2, 2]$ correspond to separable (unentangled) configurations. Values in $(2, 2\sqrt{2}]$ and $[-2\sqrt{2}, -2)$ correspond to entangled, Bell-violating configurations.

The fact that Phase Theory, like quantum mechanics, achieves exactly the Tsirelson bound and no more is a deep structural alignment between the two frameworks. The Tsirelson bound is derived in quantum mechanics from the algebra of Hilbert space operators; in Phase Theory, it is derived from the geometry of the admissible configuration space. The agreement between these two derivations is non-trivial and constitutes strong structural evidence that Phase Theory is capturing genuine features of physical reality at a deeper level than the quantum mechanical formalism.

37. Experimental Extensions and Falsifiable Predictions

Phase Theory makes eight classes of specific, falsifiable predictions regarding Bell-type experiments. These predictions are either identical to standard quantum mechanical predictions (in which case experimental confirmation of QM constitutes support for Phase Theory) or genuinely novel (in which case experiments can discriminate between Phase Theory and all other frameworks).

5. **No CHSH violation exceeding $2\sqrt{2}$ in any experiment.** Phase Theory, like quantum mechanics, predicts that no experiment will observe $|S| > 2\sqrt{2}$. Any observation of super-quantum correlations

would falsify Phase Theory. This prediction is identical to quantum mechanics and is supported by all existing data.

6. **Exact cosine correlation law for all photon polarization measurements.** Phase Theory predicts $E(\theta_1, \theta_2) = -\cos(\theta_1 - \theta_2)$ to arbitrary precision for all angles θ_1, θ_2 and all entangled photon states. This is identical to quantum mechanics and is strongly supported by existing experiments.
7. **No super-quantum correlations in any physical system.** In particular, no system of any complexity — macroscopic or microscopic — will achieve $|S| > 2\sqrt{2}$. This includes hypothetical strongly interacting condensed matter systems that have been proposed as potential sources of super-quantum correlations [28].
8. **Specific contextuality patterns beyond KS.** Phase Theory predicts that the contextuality structure of multi-observable experiments will follow the pattern dictated by the admissibility geometry, which makes specific predictions about which contextual inequalities are violated and by how much. These predictions generalize KS contextuality and constitute a novel experimental signature of Phase Theory that is not predicted by quantum mechanics alone, since quantum mechanics is silent on why the KS structure takes the form it does.
9. **No Bell-type deviations in the mesoscopic regime.** Unlike GRW, Phase Theory predicts exact quantum mechanical Bell violation rates at all scales. Precision tests of Bell violations with mesoscopic systems ($N \sim 10^3$ – 10^6 particles) should find no deviations from the quantum mechanical prediction. Observation of GRW-predicted deviations would falsify Phase Theory.

10. **Loophole-free Bell violations agree precisely with QM to**

arbitrary precision. Phase Theory predicts that future loophole-free experiments with improved precision will continue to find exact agreement with quantum mechanical predictions. There are no free parameters to tune: the prediction is determined by the $U(1)$ topology of the phase substrate and the admissibility overlap integral formula (Eq. 8).

11. **Topology-dependent correlation signatures in multi-particle (N**

> 2) entanglement experiments. For $N \geq 3$ particles, Phase Theory predicts specific correlation patterns determined by the higher homotopy groups $\pi_k(M_A)$ for $k \geq 2$. For three-particle (GHZ-type) entanglement, the winding number structure is determined by $\pi_2(M_A)$, which predicts specific relationships between the three-party correlation functions that are not automatically required by quantum mechanics alone. Experimental protocol: measure three-photon GHZ correlations as a function of simultaneous rotation of all three polarizers; the Phase Theory prediction is a specific functional form $E(\theta_1, \theta_2, \theta_3) = f(\theta_1 + \theta_2 + \theta_3)$ dictated by the admissibility functional on π_2 . This prediction is currently experimentally untested in the requisite precision.

12. **Phase-winding signatures in annihilation channel correlations.**

As established in Paper 15 of this series [PT-15], the annihilation channel of particle-antiparticle pairs carries specific phase-winding information. Phase Theory predicts that Bell tests performed on photons from positron-electron annihilation events will show specific departures from the standard cosine correlation in the angular distribution — departures arising from the non-trivial winding numbers of the annihilation channel phase structure. Experimental

protocol: measure two-photon polarization correlations from positronium annihilation events at varying photon energies; the Phase Theory prediction is a small energy-dependent correction $\delta E(\theta, E) = \alpha \cdot \sin(2\theta) \cdot (E/m_e c^2)^2 \cdot \kappa_{\text{ann}}$ where κ_{ann} is the annihilation channel winding number computed in Paper 15. This constitutes a cross-prediction between the Bell sector and the particle physics sector of Phase Theory.

Predictions 4, 7, and 8 are the most experimentally distinctive. Prediction 4 requires precision contextuality measurements beyond current state of the art. Prediction 7 requires three-photon GHZ correlation precision beyond current experiments. Prediction 8 requires precision polarimetry of positronium annihilation photons at varying energies. All three are achievable with near-future technology.

38. Implications for Quantum Information Theory

Bell inequality violations underpin some of the most important protocols in quantum information theory: quantum key distribution (QKD), quantum teleportation, entanglement-based cryptographic protocols, and device-independent quantum computation. Phase Theory's reinterpretation of Bell violations has direct implications for the foundations of all of these protocols.

Proposition 38.1 (Security of Device-Independent QKD).

The security of device-independent quantum key distribution protocols is guaranteed within Phase Theory by two results: (1) the no-signaling theorem (Theorem 17.1), which ensures that an eavesdropper cannot extract information from the quantum channel by exploiting correlations; and (2) the impossibility of factorizable eavesdropping

strategies — any eavesdropping strategy that creates a factorizable joint distribution over Alice's, Bob's, and Eve's outcomes must satisfy Bell inequalities, while the legitimate channel violates them, making eavesdropping detectable.

Proposition 38.2 (Quantum Teleportation as Admissibility-Enforced Transfer).

Quantum teleportation is re-interpreted in Phase Theory as admissibility-enforced state transfer. The classical communication channel between Alice and Bob does not transmit quantum information: it specifies which sector of the joint admissible configuration space is relevant, allowing Bob to identify which admissible configuration has been selected for his site. The "quantum information" is not transmitted superluminally; it is already encoded in the global admissible configuration.

Theorem 38.1 (All Bell-Exploiting QI Protocols Realizable Without Nonlocality).

All quantum information protocols that exploit Bell inequality violations — including QKD, teleportation, superdense coding, and device-independent protocols — are fully realizable within Phase Theory without invoking nonlocal causation. The protocols work because they exploit the admissibility constraints on the joint configuration space, not because they transmit information faster than light.

The practical implication of Theorem 38.1 is that quantum information technology is, from the Phase Theory perspective, not "quantum" in the sense of invoking mysterious quantum principles, but is rather the technology of exploiting global admissibility constraints. The engineering challenge of building quantum computers and quantum communication systems is the challenge of creating and maintaining globally admissible phase configurations — a challenge that Phase Theory suggests may have a geometric formulation in terms of maintaining non-trivial topological correlation indices $\kappa(\varphi)$ against decoherence.

39. Implications for Quantum Gravity

The relationship between quantum mechanics and general relativity is one of the deepest unsolved problems in physics. Bell violations create a specific tension within this broader problem: if the correlations implied by Bell inequality violations are genuinely nonlocal — if they involve some form of superluminal influence — then they seem to conflict with the local structure of spacetime that general relativity presupposes. This tension is at the heart of several proposals for quantum gravity, including holographic approaches and ER=EPR [29].

Proposition 39.1 (No Tension with General Relativity in Phase Theory).

In Phase Theory, Bell violations are topological, not causal. They do not involve any superluminal influence between spacelike-separated events. Therefore there is no tension between Bell violations and the local causal structure of spacetime as described by general relativity.

Theorem 39.1 (Bell Violations and Gravity Share the Same Functional).

The admissibility functional $A[\varphi]$ that governs Bell inequality violations (by constraining the joint outcome distributions) is the same functional that governs gravitational dynamics in the classical limit (as established in Paper 5 [PT-5]). Gravity and Bell violations are not competing structures — they are two faces of the same global admissibility structure at different limits.

Proof sketch. In Paper 5, the Einstein field equations are derived as the classical (low-winding-number, high-amplitude) limit of the admissibility stationarity conditions $\delta A[\varphi]/\delta\varphi = 0$. The same stationarity conditions, applied to high-winding-number configurations appropriate for entangled microscopic systems, yield the Bell-violating correlation functions of Sections 13–22. The single functional $A[\varphi]$ generates both sets of results under different limiting conditions. $\square$

Corollary 39.1 (Unified Account of Gravity and Bell Violations).

Phase Theory provides a unified account in which quantum gravity and Bell inequality violation are two aspects of the same global admissibility structure. Quantum gravity is the regime in which gravitational and entanglement contributions to $A[\varphi]$ become comparable; Bell inequality violations are the fingerprint of this global structure at the level of individual quantum systems.

This unification is one of the most promising aspects of Phase Theory as a research program. The longstanding conflict between quantum mechanics and general relativity — which takes many forms but is at its deepest level a

conflict between the global structure of quantum entanglement and the local causal structure of spacetime — is dissolved in Phase Theory by recognizing that both are aspects of a single global admissibility structure. The ER=EPR conjecture [29], which proposes that entanglement and wormholes are related by the same underlying physics, is a suggestive precursor to the Phase Theory unification, though the precise relationship between these frameworks is a subject for future investigation.

40. The Measurement Problem Revisited

The *measurement problem* is the deepest conceptual problem in the foundations of quantum mechanics: why does measurement produce a definite outcome? Quantum mechanics describes systems by superpositions of possibilities; measurement somehow selects a single definite outcome. Without additional postulates (collapse) or additional ontology (branching worlds or hidden variables), this transition from superposition to definite outcome is unexplained.

Bell inequality violations add a second layer to this problem: not only must we explain why measurements produce definite outcomes, but we must also explain how the outcomes at distant sites are correlated. The two problems are logically related: if measurement produces a definite outcome through some nonlocal mechanism, then perhaps the same mechanism also enforces the correlations; if the correlations are structural, perhaps the definiteness of outcomes is also structural.

Phase Theory resolves both problems simultaneously through global admissibility.

Theorem 40.1 (Simultaneous Resolution of Measurement and Bell

Problems).

The Phase Theory resolution of the measurement problem (definiteness of outcomes follows from admissibility enforcement) simultaneously resolves the Bell correlation problem (inter-site correlations are features of the global admissible configuration). Both problems dissolve when outcomes are understood as consistent selections from globally admissible phase configurations rather than collapses of wave functions or realizations in branches.

Proof. Definiteness of outcomes: An "outcome" in Phase Theory is not a random selection from a probability distribution but the identification, by the measurement apparatus, of which admissible configuration has been selected by the global admissibility enforcer. The configuration is definite; the outcome is definite. There is no collapse because there was no superposition in the fundamental ontology — there was only one globally admissible configuration, which the measurement reveals. Correlation of outcomes: The globally admissible configuration selected by admissibility enforcement assigns consistent outcomes to all measurement contexts simultaneously (Definition 10.1). The correlation between site A and site B outcomes is a feature of the single globally realized configuration, not a result of any communication or influence between the sites. Both problems are resolved by the same mechanism: global admissibility selects a single, globally consistent configuration, and measurements reveal aspects of this configuration. $\square$

The resolution is parsimonious: no collapse mechanism, no branching ontology, no hidden variables, no nonlocal influence. Only a single globally admissible phase configuration that is consistent across all measurement contexts simultaneously. The apparent randomness of quantum mechanics —

the fact that we cannot predict which outcome will occur at any given measurement — is explained in Phase Theory as an epistemological limitation: we do not have access to the full global configuration, only to the local aspect of it that the measurement probes.

41. Relationship to Other Papers in the Phase Theory Series

This paper is Paper 32 of the Phase Theory series and builds directly on several preceding papers while setting up the following ones.

Paper 1: Foundations and Axioms [PT-1]. The four axioms PT-1 through PT-4 stated in Section 7 of this paper are first introduced and motivated in Paper 1. The admissibility functional $A[\varphi]$ and the admissible configuration space M_A are defined there in full generality. The present paper applies those foundations to the specific problem of Bell inequality violations.

Paper 5: Spacetime Emergence [PT-5]. Paper 5 derives the emergence of spacetime geometry from admissibility constraints and establishes that Einstein's field equations arise as the classical limit of the admissibility stationarity conditions. Theorem 39.1 of the present paper relies on this result to establish that gravity and Bell violations share the same fundamental functional.

Paper 15: Particle Spectrum and Annihilation Channels [PT-15]. Paper 15 establishes the Phase Theory account of the particle spectrum, including the computation of annihilation channel winding numbers κ_{ann} . Falsifiable prediction 8 of Section 37 is a cross-prediction between the Bell sector and the particle physics sector, relying on the annihilation channel analysis of Paper 15.

Paper 29: Entanglement in Phase Theory [PT-29]. Paper 29 provides the full Phase Theory account of quantum entanglement, establishing the topological characterization of entangled states and the relationship between the entanglement entropy and the winding number structure of the admissible configuration space. The present paper uses these results (especially the definition of entanglement via non-product admissible spaces) extensively in Sections 23–24.

Paper 31: No-Signaling Theorem [PT-31]. Paper 31 provides the full no-signaling theorem in the most general formulation within Phase Theory. The present paper provides a self-contained proof for the Bell scenario (Theorem 17.1) and refers to Paper 31 for the most general result.

Paper 33: Contextuality (following paper). Paper 33, the paper immediately following this one in the series, will develop the full account of quantum contextuality in Phase Theory, extending the results of Sections 18–19 of the present paper to the general multi-observable case. The present paper establishes that contextuality is structurally necessary (Theorem 18.1) and that Phase Theory satisfies KS conditions (Theorem 19.1); Paper 33 will develop the quantitative contextuality theory.

The overall structure of the Phase Theory program is a progressive development from the four foundational axioms (Paper 1) through the recovery of spacetime (Paper 5), particle physics (Papers 10–20), quantum mechanics (Papers 25–35), and ultimately quantum gravity (Papers 36–50). The present paper occupies a central position in the quantum mechanics sector, addressing the most fundamental challenge to any proposed unifying framework: the explanation of Bell inequality violations without nonlocal causation.

42. Discussion: Open Questions

The framework developed in this paper, while complete at the level of conceptual and mathematical structure, leaves several important questions open for future investigation.

Formal Derivation of the Admissibility Functional from First Principles. The admissibility functional $A[\varphi]$ has been postulated (Axiom PT-2) and its consequences derived. A fundamental open question is whether $A[\varphi]$ can be derived from more primitive considerations — specifically, whether the requirement of global consistency of a phase-bearing substrate uniquely determines the form of $A[\varphi]$. Preliminary results suggest that $A[\varphi]$ is the unique functional consistent with $U(1)$ gauge invariance, diffeomorphism invariance, and the topological constraints of the phase substrate, but a complete proof of uniqueness remains to be provided.

Computation of Phase Winding Numbers for Three- and Four-Party Entanglement. The present paper treats two-party entanglement and the CHSH scenario. The extension to three-party (GHZ-type) and four-party entanglement requires computing the relevant winding numbers $w(\varphi, \gamma_{ABC})$ and $w(\varphi, \gamma_{ABCD})$ for loops encircling three and four measurement regions respectively. These winding numbers are determined by the higher homotopy groups $\pi_2(M_A)$ and $\pi_3(M_A)$, whose computation is substantially more involved than the $\pi_1(M_A)$ case. This is the subject of Paper 33 and ongoing research.

Extension to Continuous-Variable Bell Inequalities. The present paper focuses on discrete (binary) outcomes. Bell inequalities for continuous outcomes — such as the position and momentum Bell inequalities proposed by Grangier et al. [30] — require a continuous-outcome analog of the admissibility framework. This extension is non-trivial because the outcome space $O = \mathbb{R}$ is non-compact, requiring modifications to the admissibility functional and the overlap integral formula (Eq. 8).

Connection Between Admissibility Geometry and Gauge Field Theory.

As discussed in Paper 18 [PT-18], the Phase Theory admissibility structure has deep connections with gauge field theory: the admissibility constraint $A[\varphi]$ resembles a gauge-invariant functional, and the admissible configuration space M_A resembles the moduli space of a gauge theory. Making this connection precise — and in particular, identifying which gauge group corresponds to the full Phase Theory admissibility structure — is an important open problem.

The Tsirelson Bound and the Topology of M_A . Theorem 14.1 derives the Tsirelson bound $2\sqrt{2}$ from the $U(1)$ structure of the phase substrate. But the specific value $2\sqrt{2}$ depends on the dimension of M_A and its curvature. A deeper question is: what property of M_A uniquely determines the Tsirelson bound at exactly $2\sqrt{2}$? Is the bound $2\sqrt{2}$ a topological invariant of M_A , or does it depend on the metric? Preliminary calculations suggest it is determined by the holonomy group ($\text{Hol}(M_A) = U(1)$ in the spin-1/2 case, $SO(3)$ in the spin-1 case) and is therefore a topological invariant, but a full proof requires more detailed analysis of the relationship between holonomy and correlation bounds.

43. Conclusion

We have demonstrated, through formal proofs and systematic analysis, that Bell inequalities fail not because the universe communicates faster than light — but because physical reality is globally organized by admissible phase structure that is irreducibly non-separable. The "mystery" of Bell inequality violation is not a mystery about nature but a consequence of mistaking the ontological ground of correlation: nature is not locally separable; it is globally phase-admissible.

The main results of this paper are:

13. Bell inequality violation is a structural necessity of global admissibility for topologically non-trivial configurations (Theorem 12.1), not a contingent feature of any particular theory.
14. The Tsirelson bound $2\sqrt{2}$ is a geometric consequence of admissibility on the $U(1)$ phase substrate (Theorem 14.1), not a quantum mechanical postulate.
15. No-signaling is a theorem, not a postulate, within Phase Theory — it follows from the gauge symmetry of the admissibility functional (Theorem 17.1).
16. The Separability Failure Theorem (Theorem 24.1) establishes that factorizability is strictly incompatible with global admissibility for entangled configurations.
17. Phase Theory reproduces all experimental Bell test results without invoking nonlocal causation, hidden variables, many-world branching, or wave function collapse.
18. Eight falsifiable predictions, including three that genuinely discriminate Phase Theory from all other frameworks, are enumerated with specific experimental protocols.

Phase Theory requires three corrected conclusions from Bell's theorem — conclusions that replace the standard three and are better supported by the evidence:

19. **Separability fails.** Physical reality is not separable into independent local states. The state of a composite system is not, in general, decomposable into the states of its components.
20. **Factorizability is not a fundamental principle.** The joint probability distribution over outcomes at spacelike-separated sites need not factor into independent local contributions. This failure is

not evidence of nonlocal causation; it is evidence of global structural organization.

21. Global admissibility is the ontological ground of correlation.

The correlations observed in Bell experiments are features of the global admissible phase configuration — topological invariants of the phase substrate that cannot be changed by local operations and cannot be exploited for superluminal signaling.

Bell did not discover that the universe communicates faster than light. He discovered that the universe is non-separable — that its physical correlations are organized globally, not locally. Phase Theory provides the framework within which this discovery can be precisely understood, mathematically formalized, and experimentally tested with predictions that go beyond quantum mechanics itself. The universe is not "nonlocal." It is globally phase-admissible. That is the correct conclusion of Bell's theorem.

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